

BFS Traversal

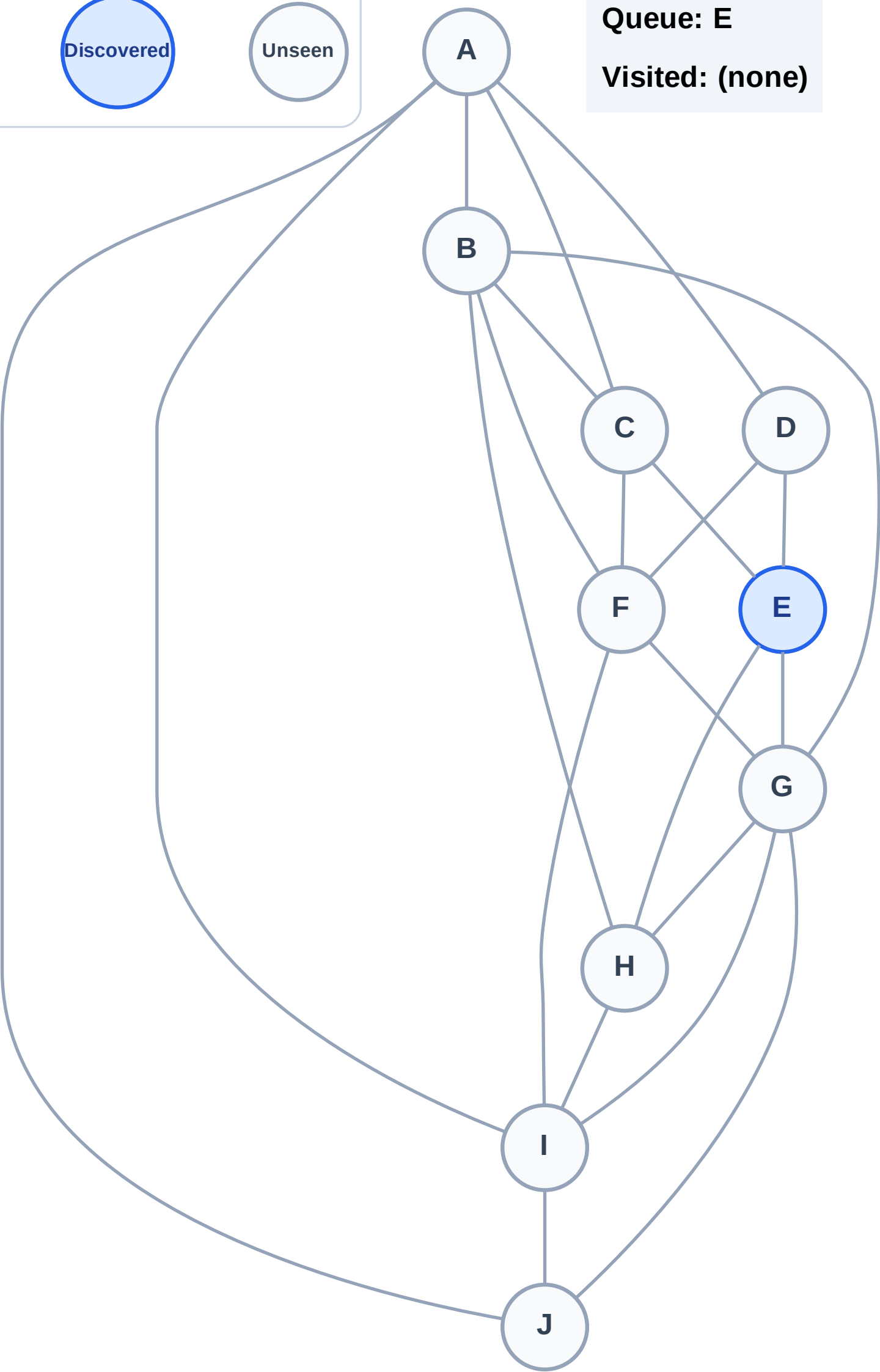
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

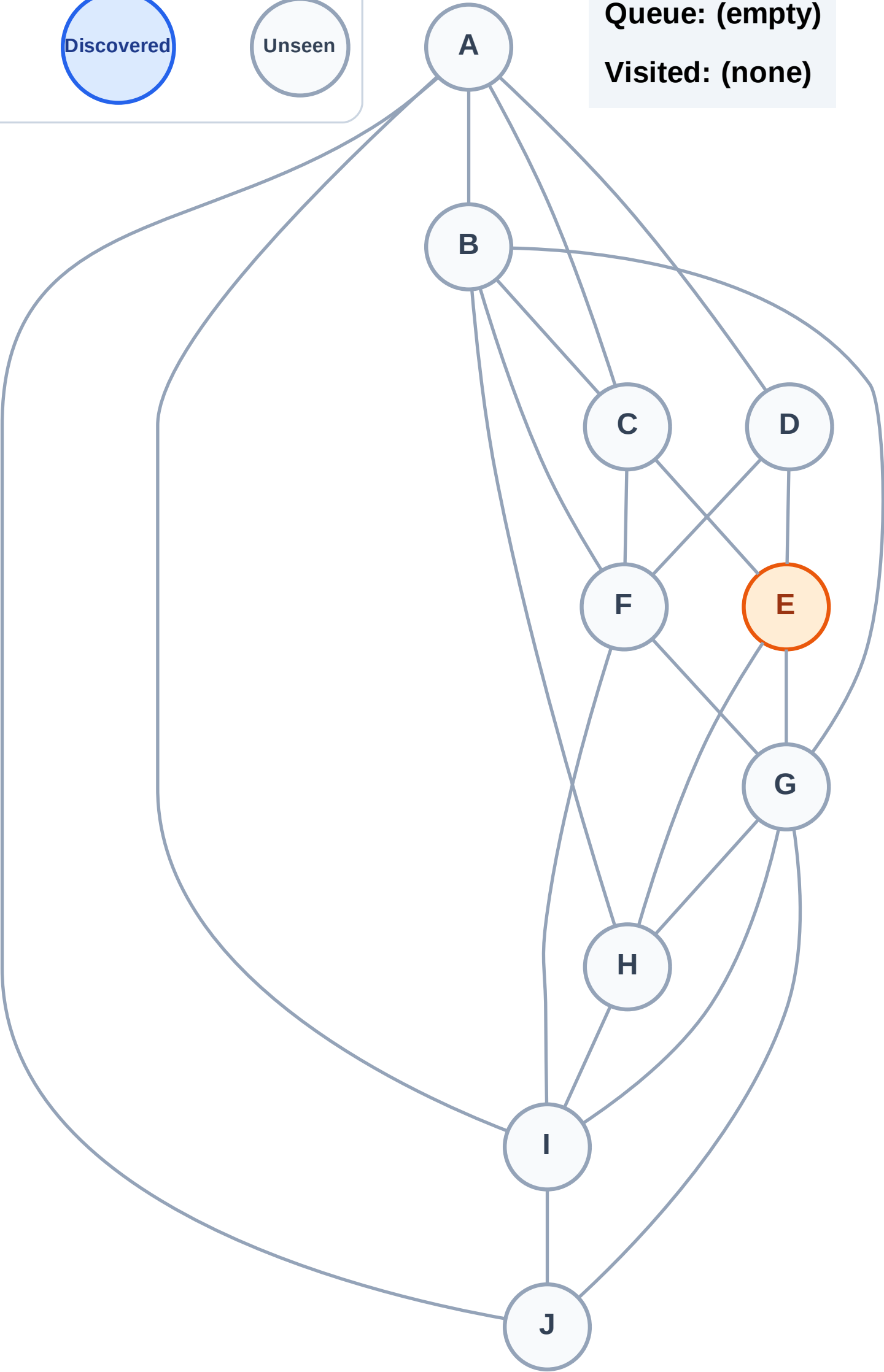
Complete

Processing

Discovered

Unseen

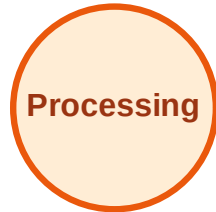
Queue: (empty)
Visited: (none)



BFS Traversal

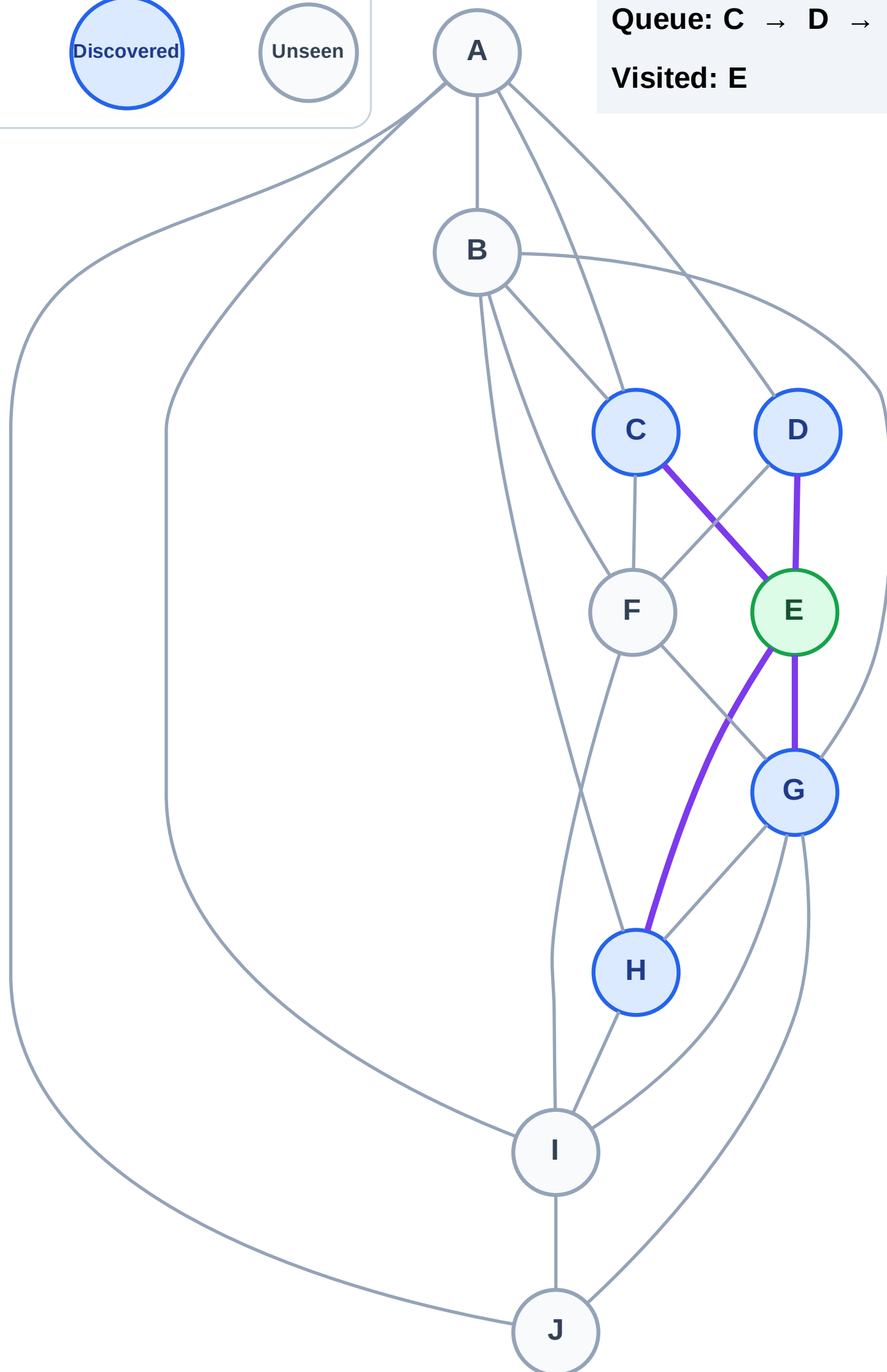
Step 3: Discover C, D, G, H from E

Legend



Queue: C → D → G → H

Visited: E



BFS Traversal

Step 4: Process node C

Legend

Complete

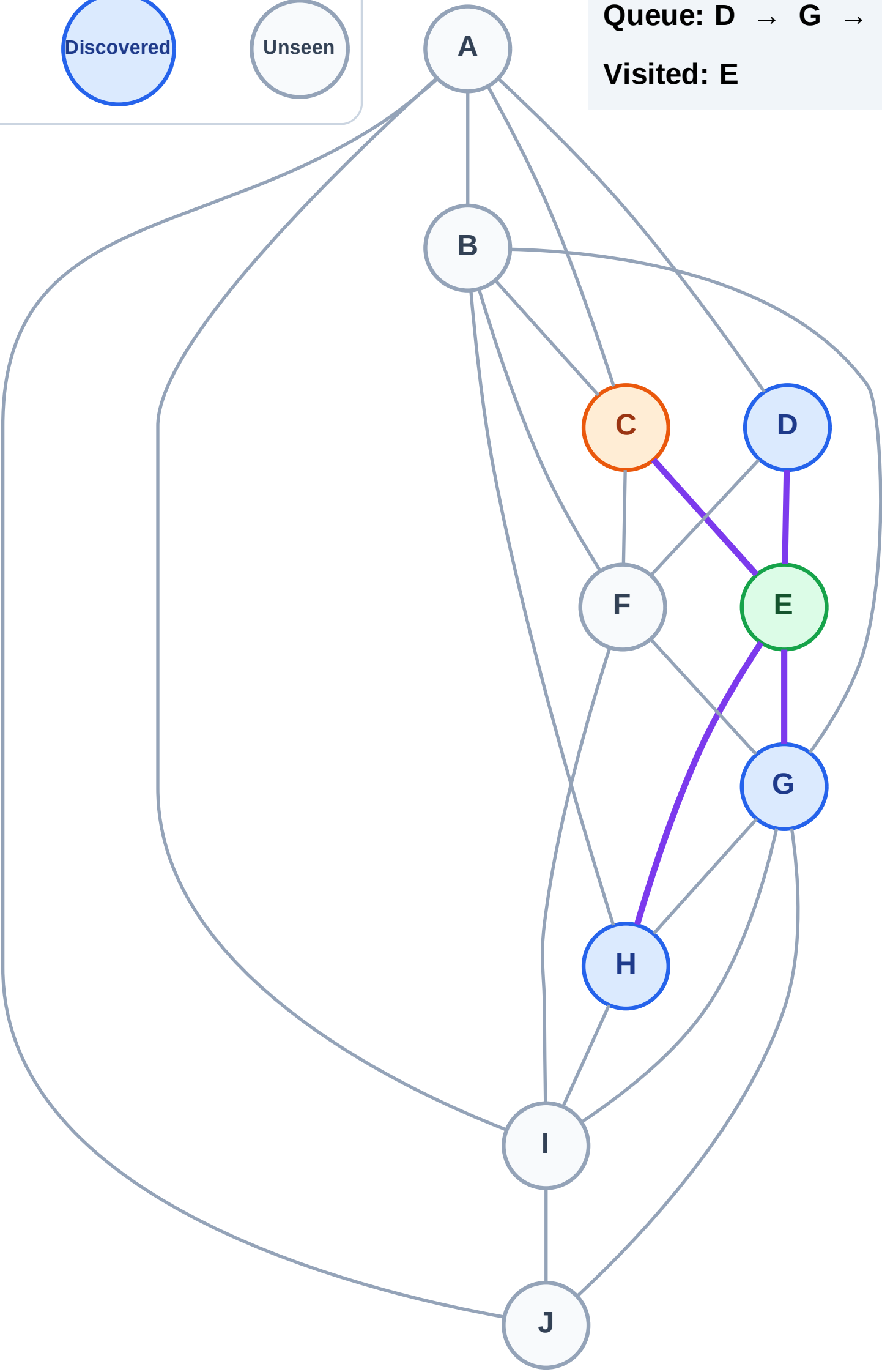
Processing

Discovered

Unseen

Queue: D → G → H

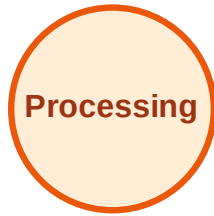
Visited: E



BFS Traversal

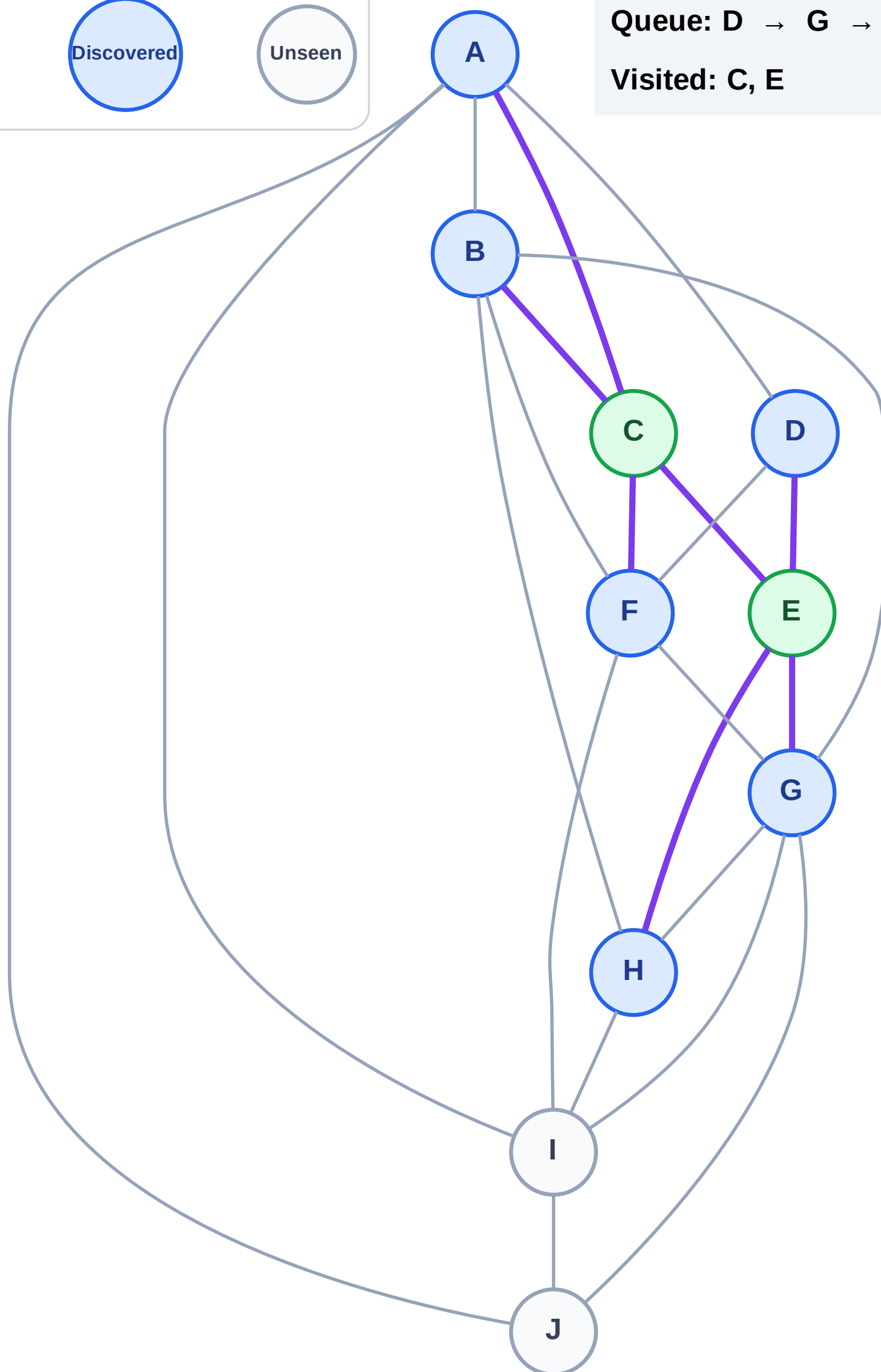
Step 5: Discover A, B, F from C

Legend



Queue: D → G → H → A → B → F

Visited: C, E



BFS Traversal

Step 6: Process node D

Legend

Complete

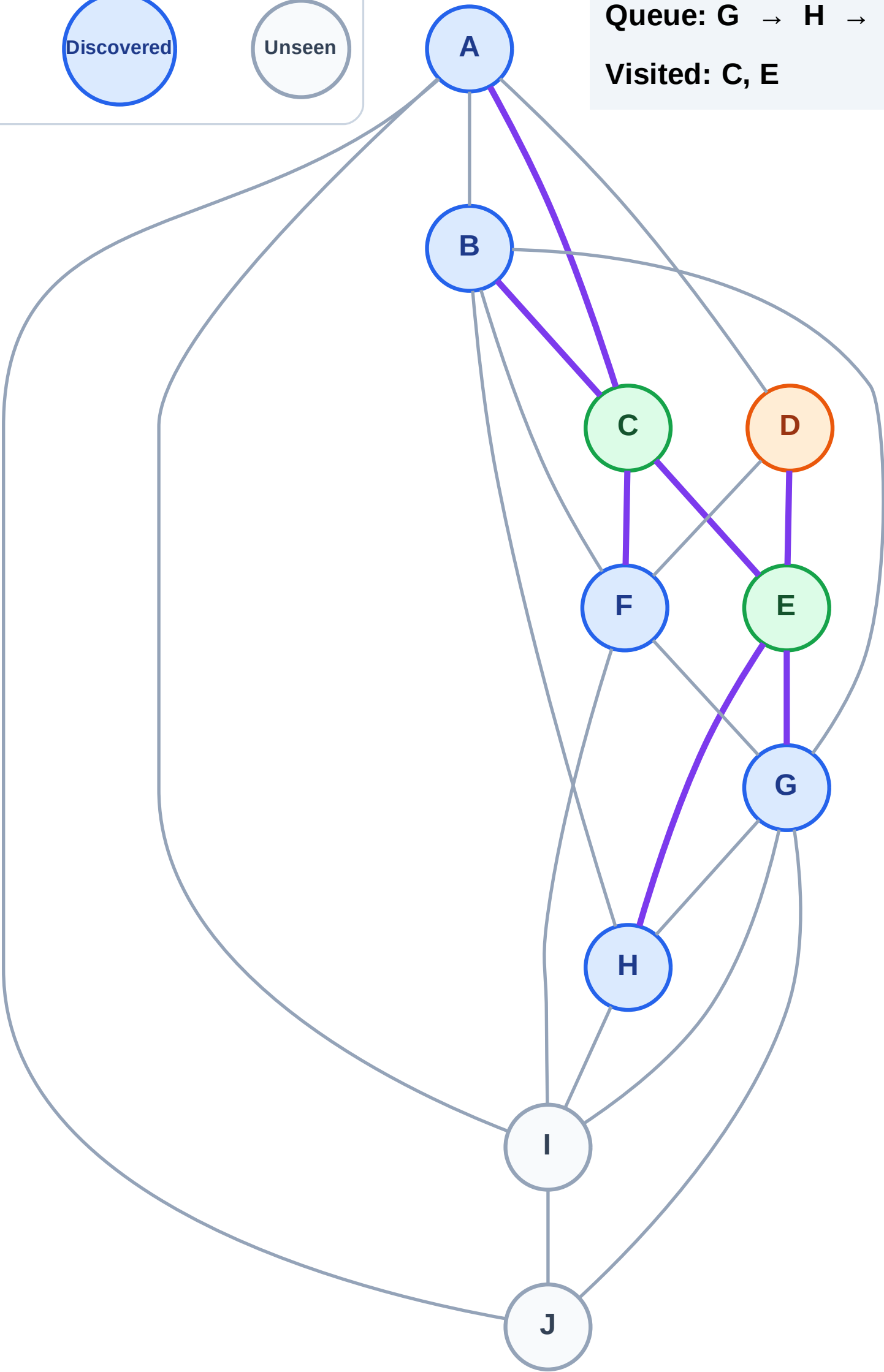
Processing

Discovered

Unseen

Queue: G → H → A → B → F

Visited: C, E



BFS Traversal

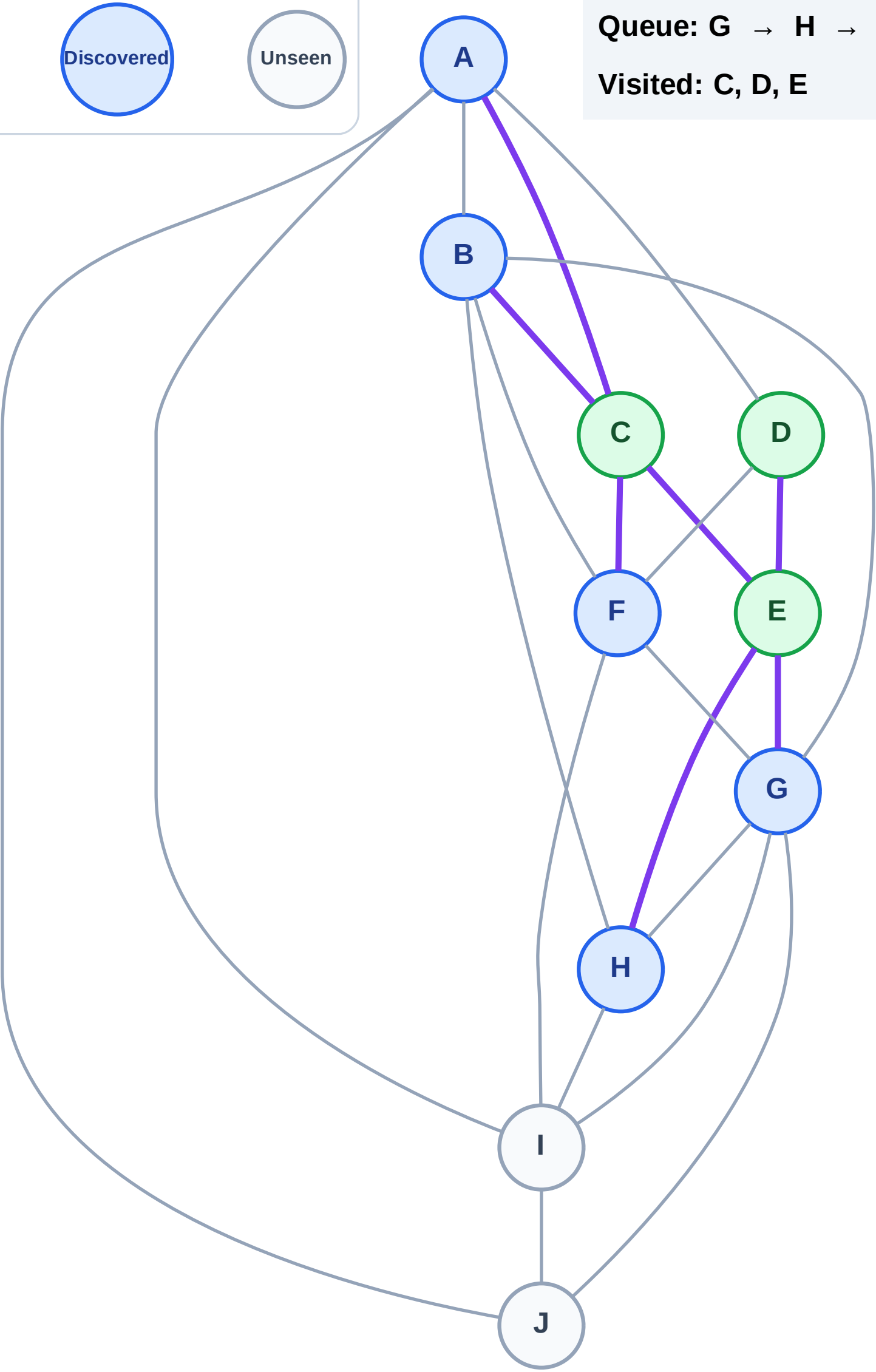
Step 7: No new neighbors from D

Legend



Queue: G → H → A → B → F

Visited: C, D, E



BFS Traversal

Step 8: Process node G

Legend

Complete

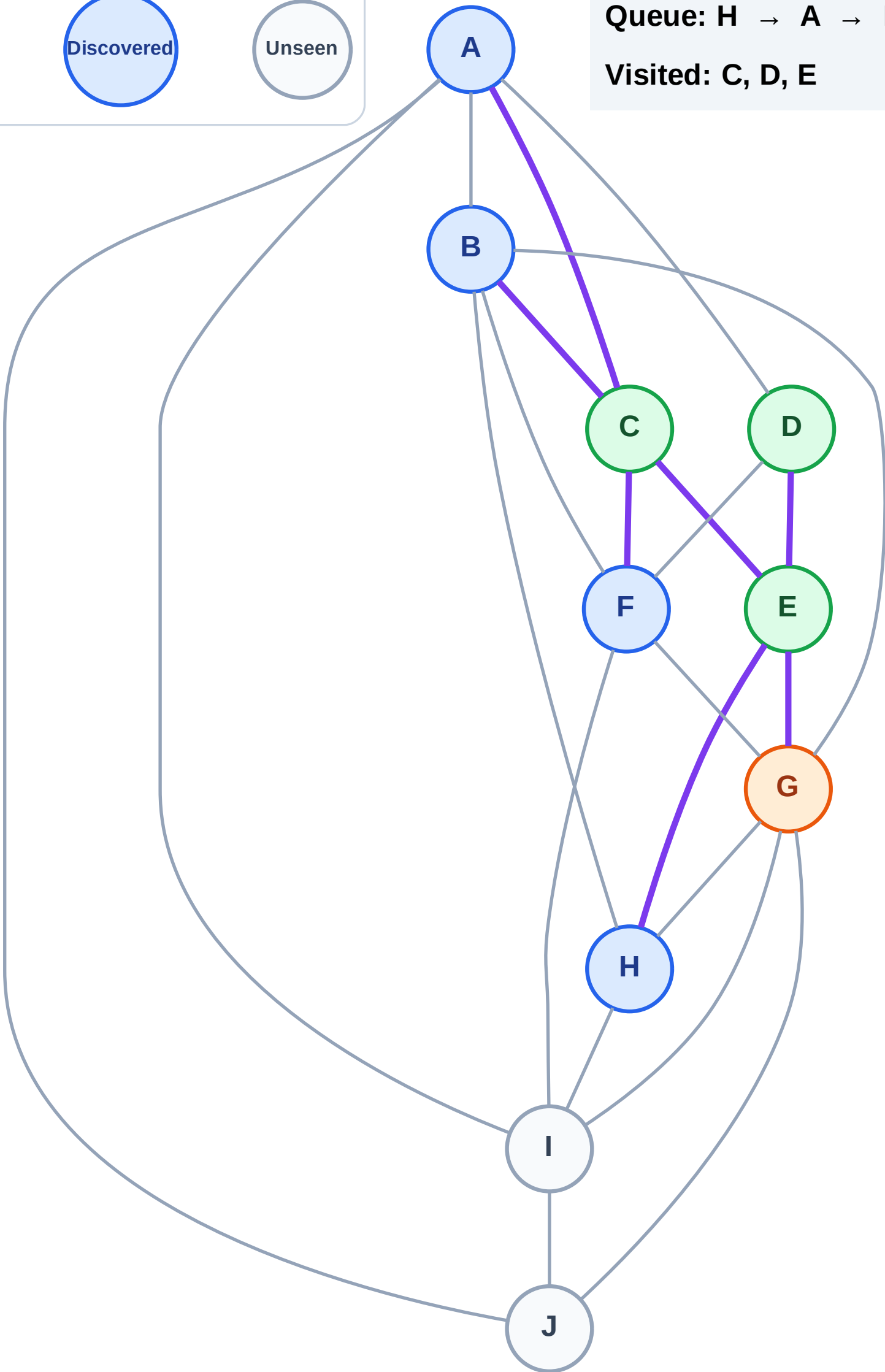
Processing

Discovered

Unseen

Queue: H → A → B → F

Visited: C, D, E



BFS Traversal

Step 9: Discover I, J from G

Legend

Complete

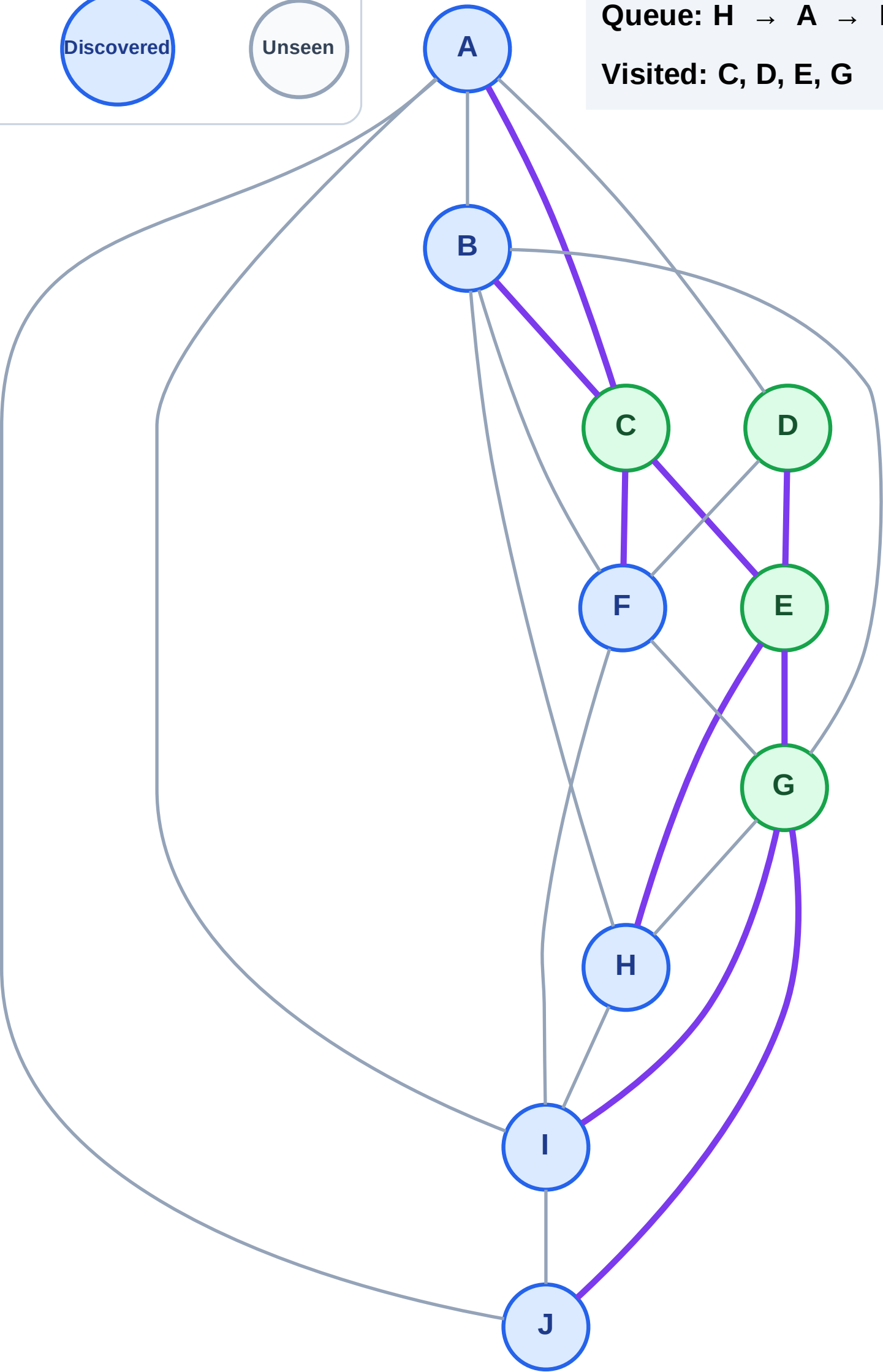
Processing

Discovered

Unseen

Queue: H → A → B → F → I → J

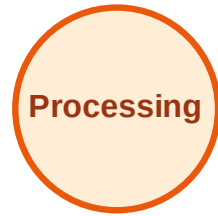
Visited: C, D, E, G



BFS Traversal

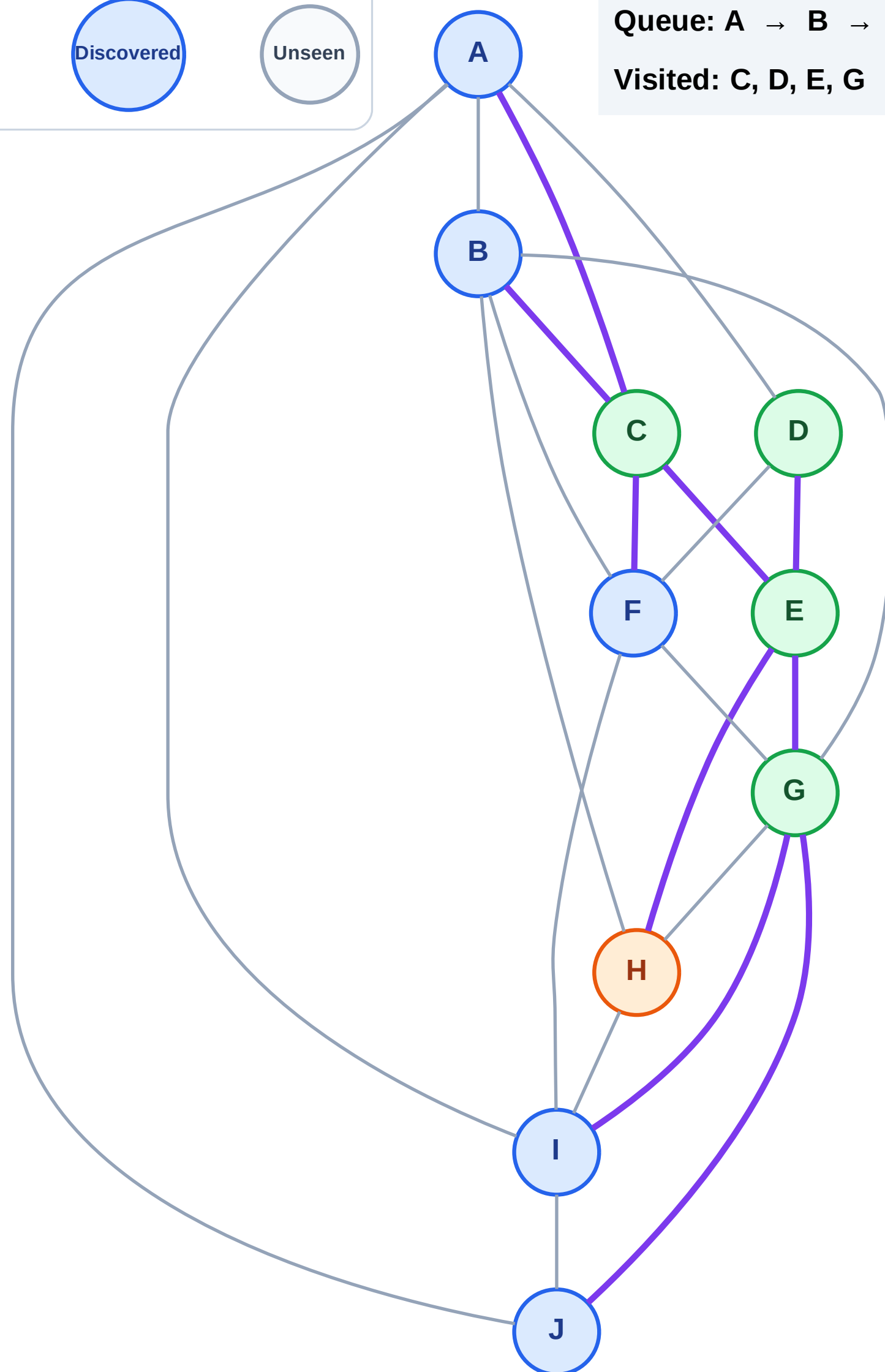
Step 10: Process node H

Legend



Queue: A → B → F → I → J

Visited: C, D, E, G



BFS Traversal

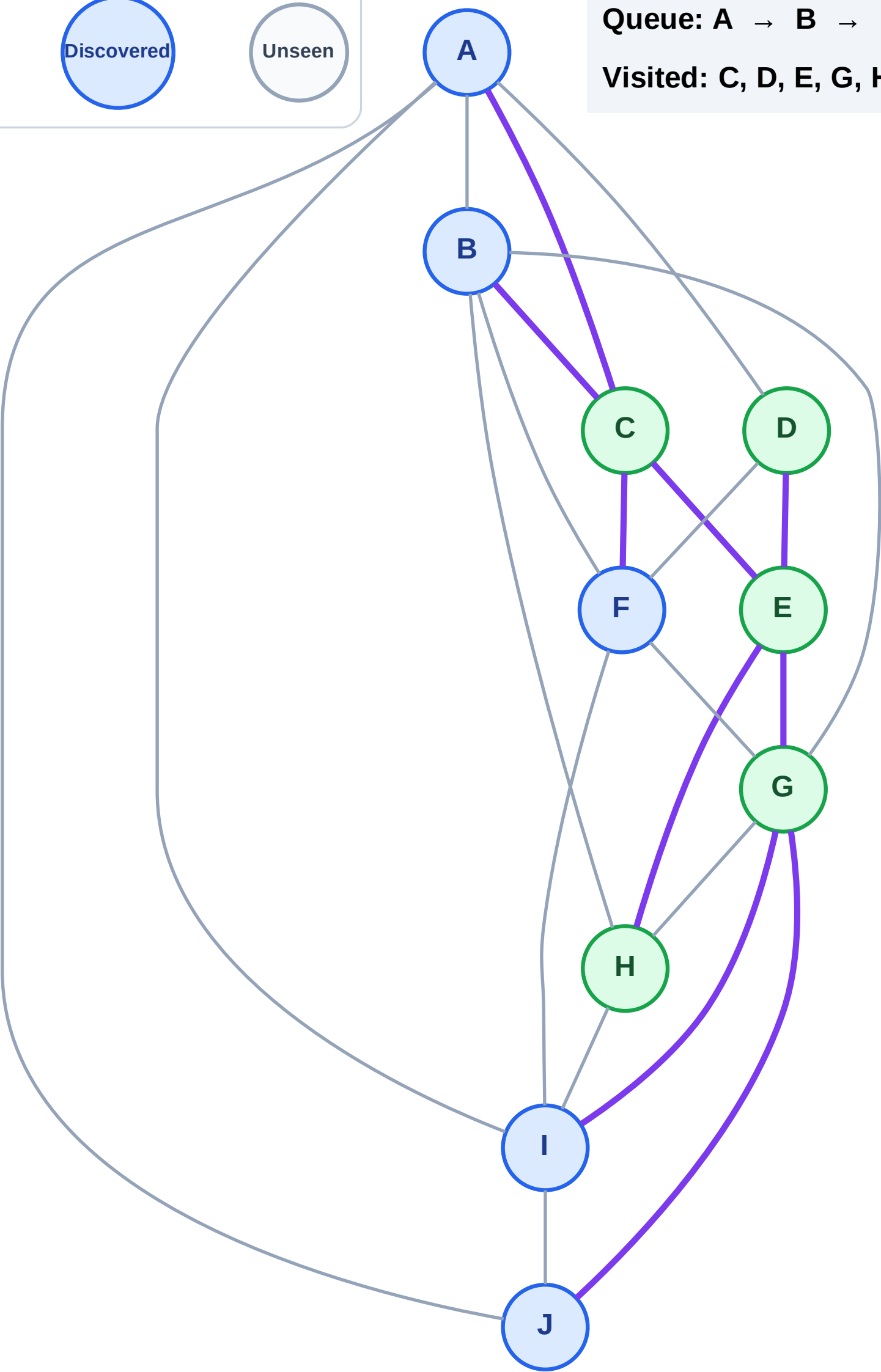
Step 11: No new neighbors from H

Legend



Queue: A → B → F → I → J

Visited: C, D, E, G, H



BFS Traversal

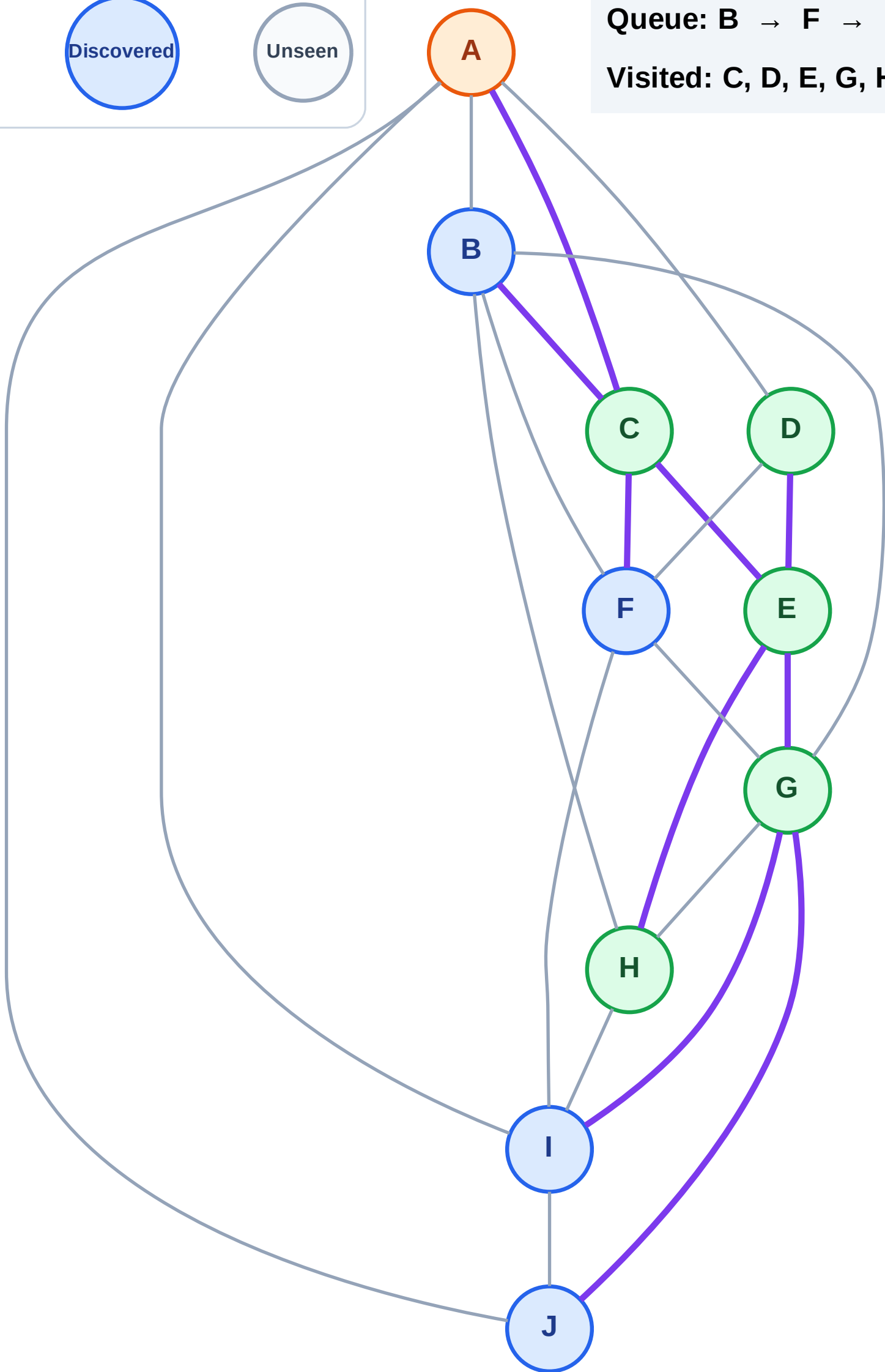
Step 12: Process node A

Legend



Queue: B → F → I → J

Visited: C, D, E, G, H



BFS Traversal

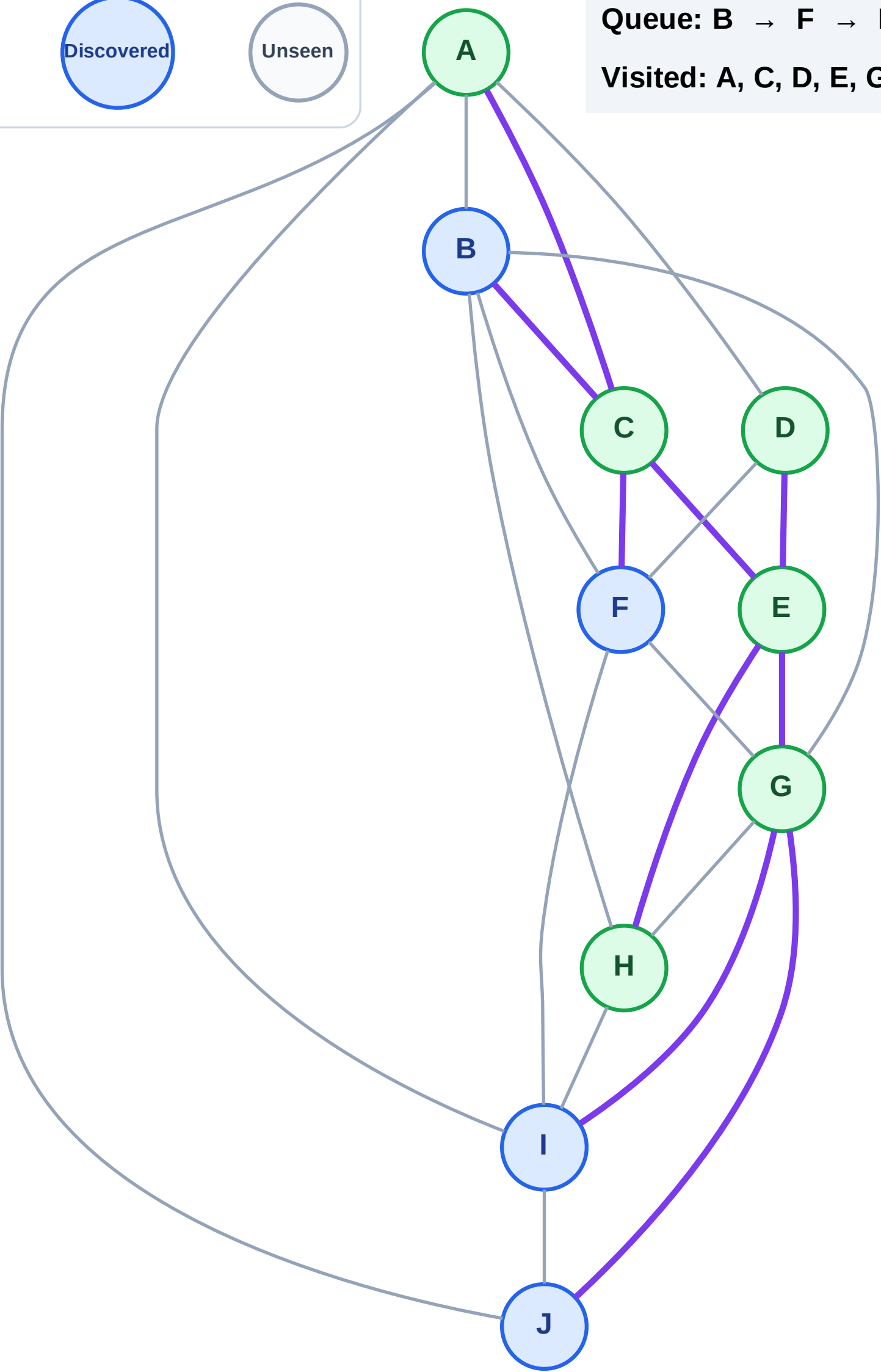
Step 13: No new neighbors from A

Legend



Queue: B → F → I → J

Visited: A, C, D, E, G, H



BFS Traversal

Step 14: Process node B

Legend

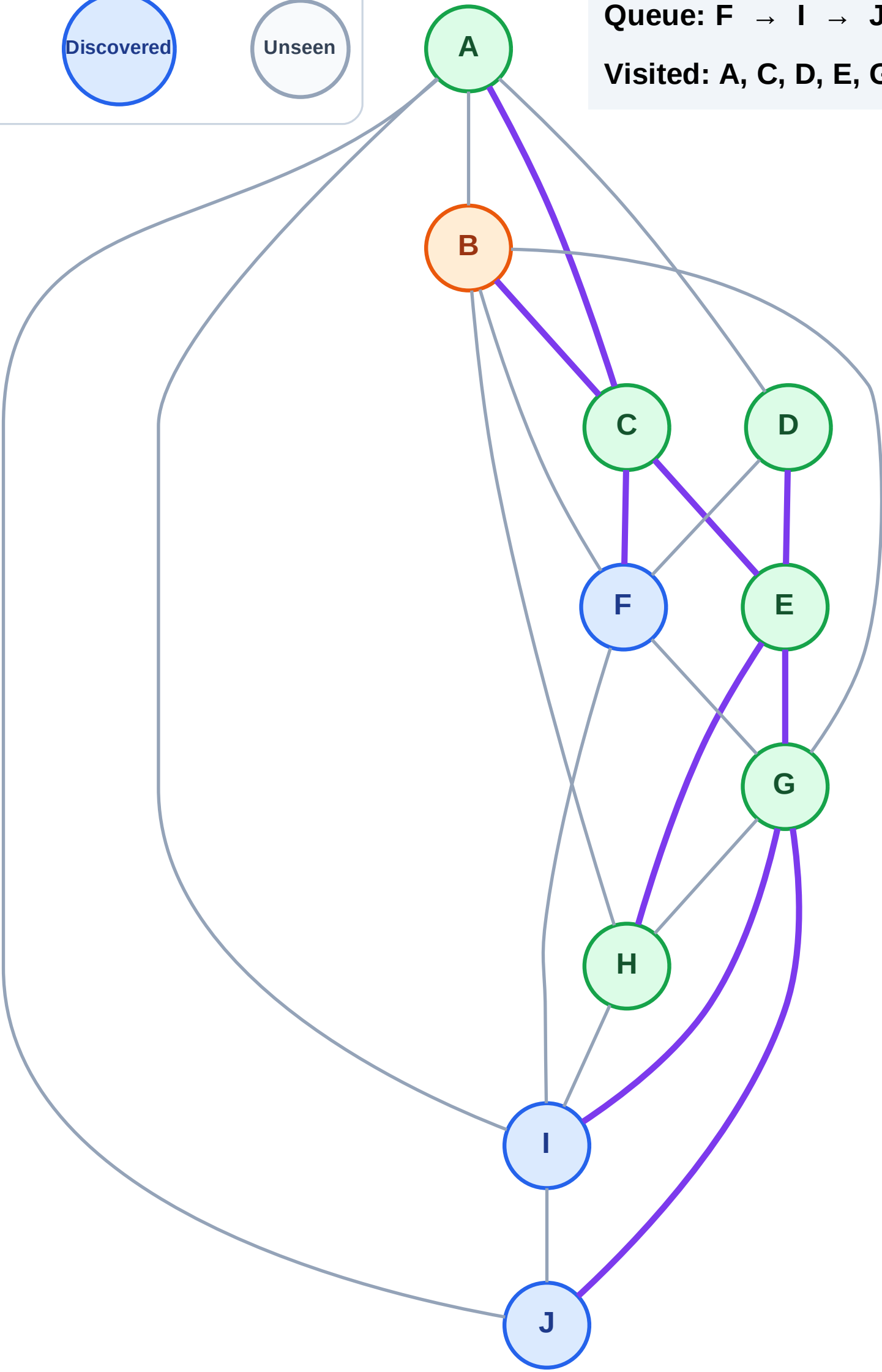
Complete

Processing

Discovered

Unseen

Queue: F → I → J
Visited: A, C, D, E, G, H



BFS Traversal

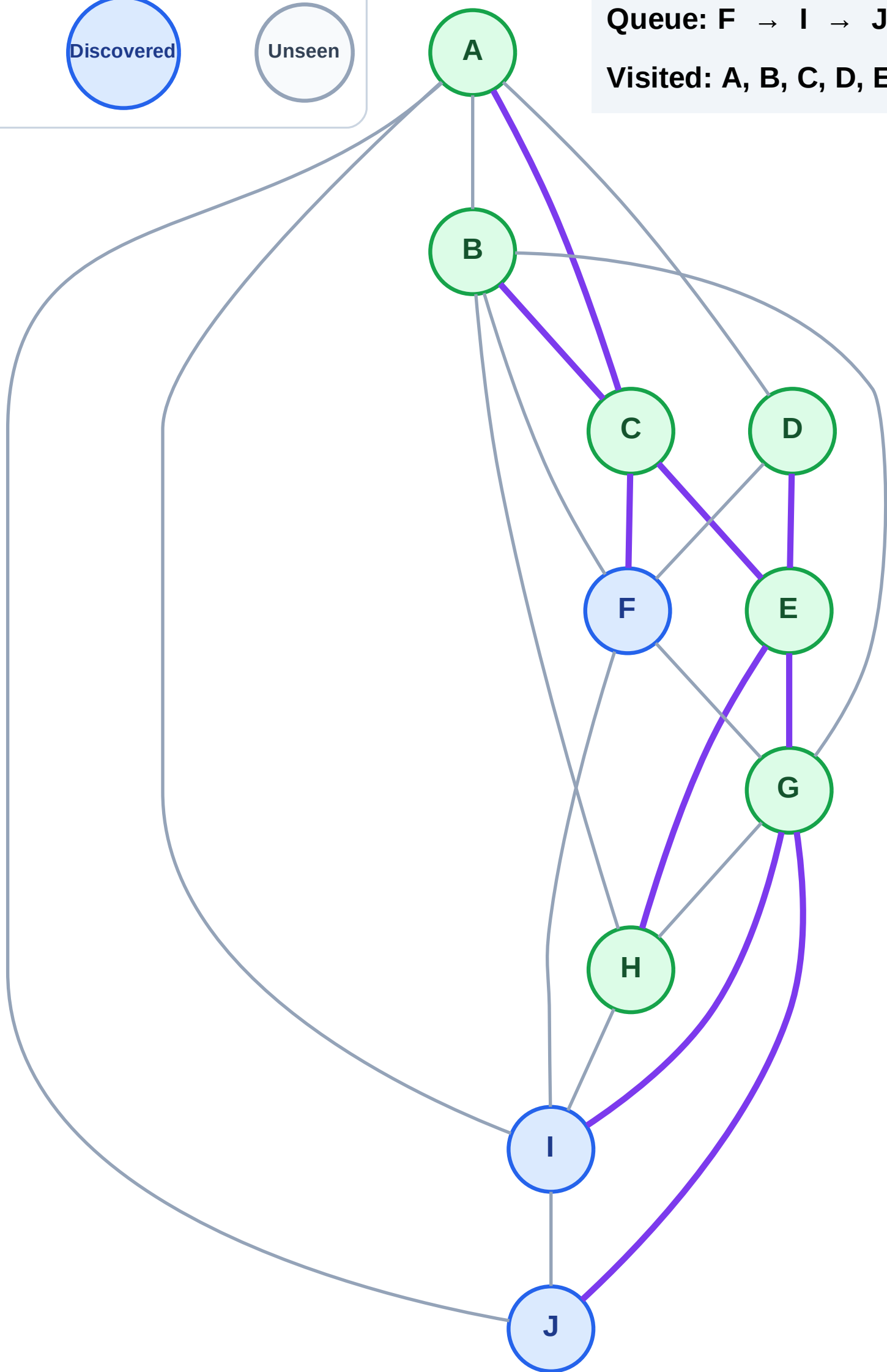
Step 15: No new neighbors from B

Legend



Queue: F → I → J

Visited: A, B, C, D, E, G, H



BFS Traversal

Step 16: Process node F

Legend

Complete

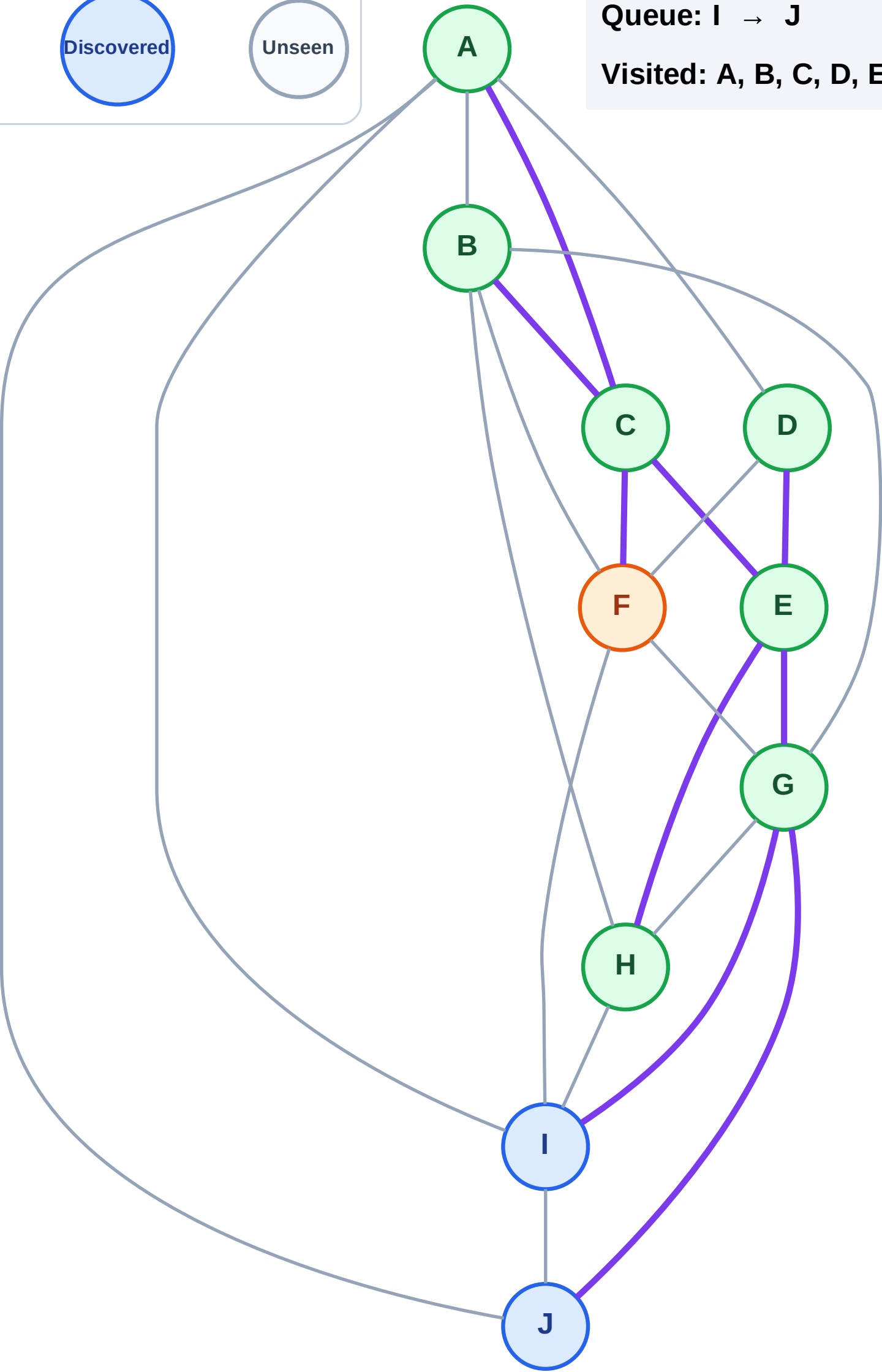
Processing

Discovered

Unseen

Queue: I → J

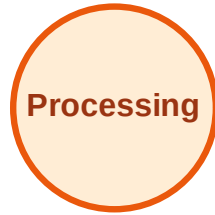
Visited: A, B, C, D, E, G, H



BFS Traversal

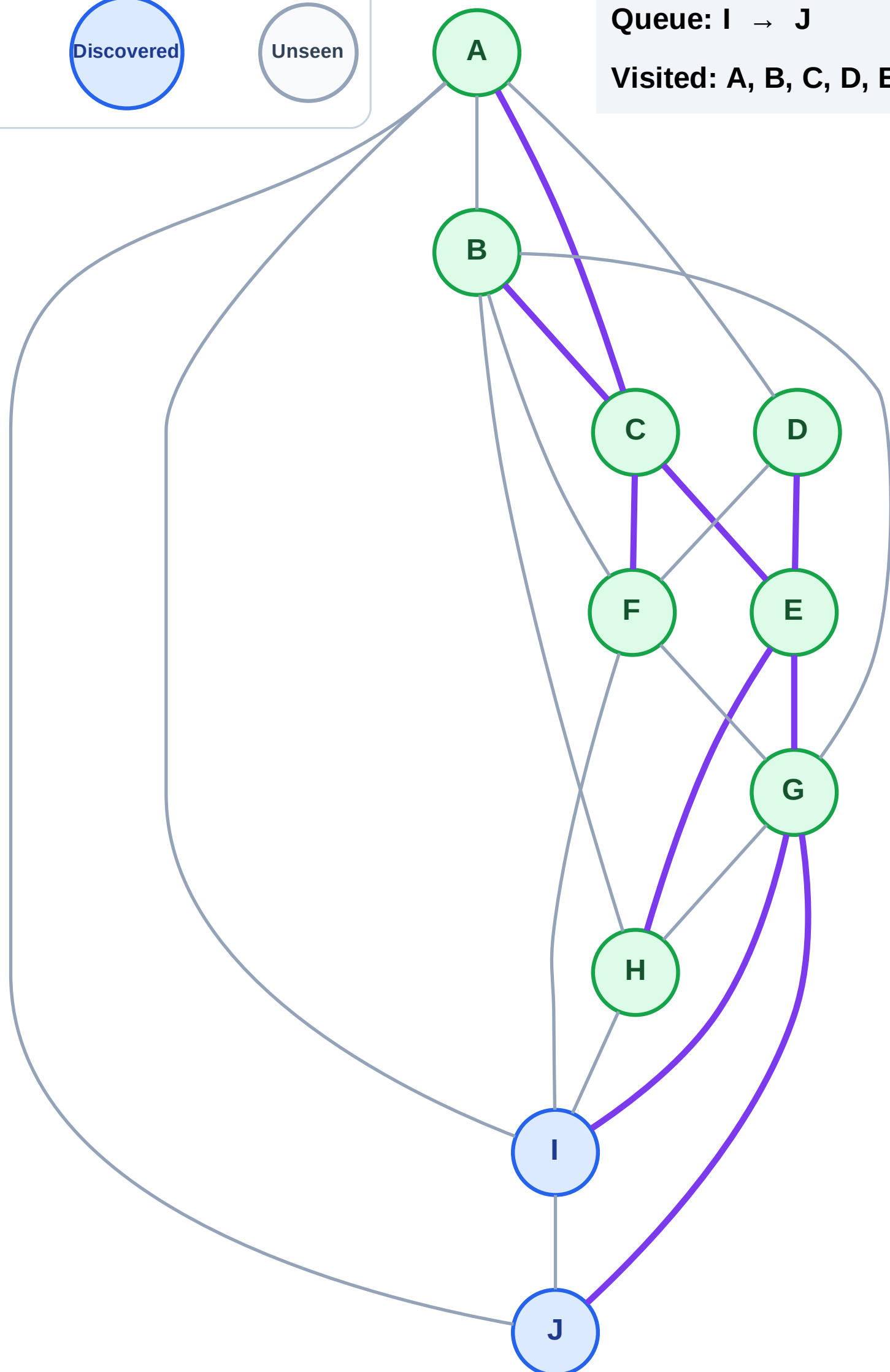
Step 17: No new neighbors from F

Legend



Queue: I → J

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node I

Legend

Complete

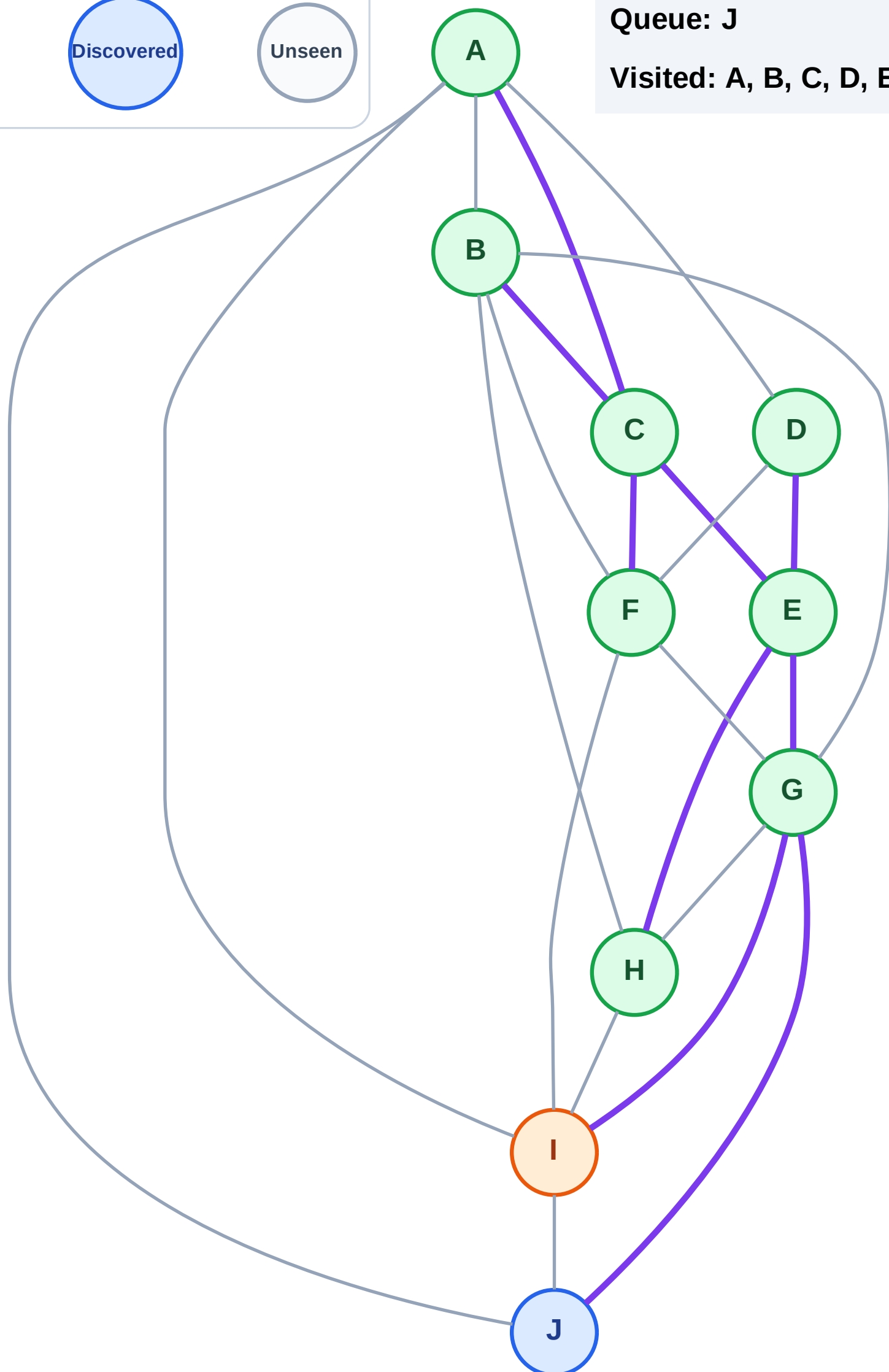
Processing

Discovered

Unseen

Queue: J

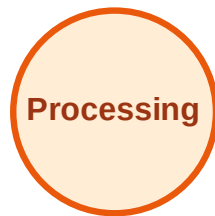
Visited: A, B, C, D, E, F, G, H



BFS Traversal

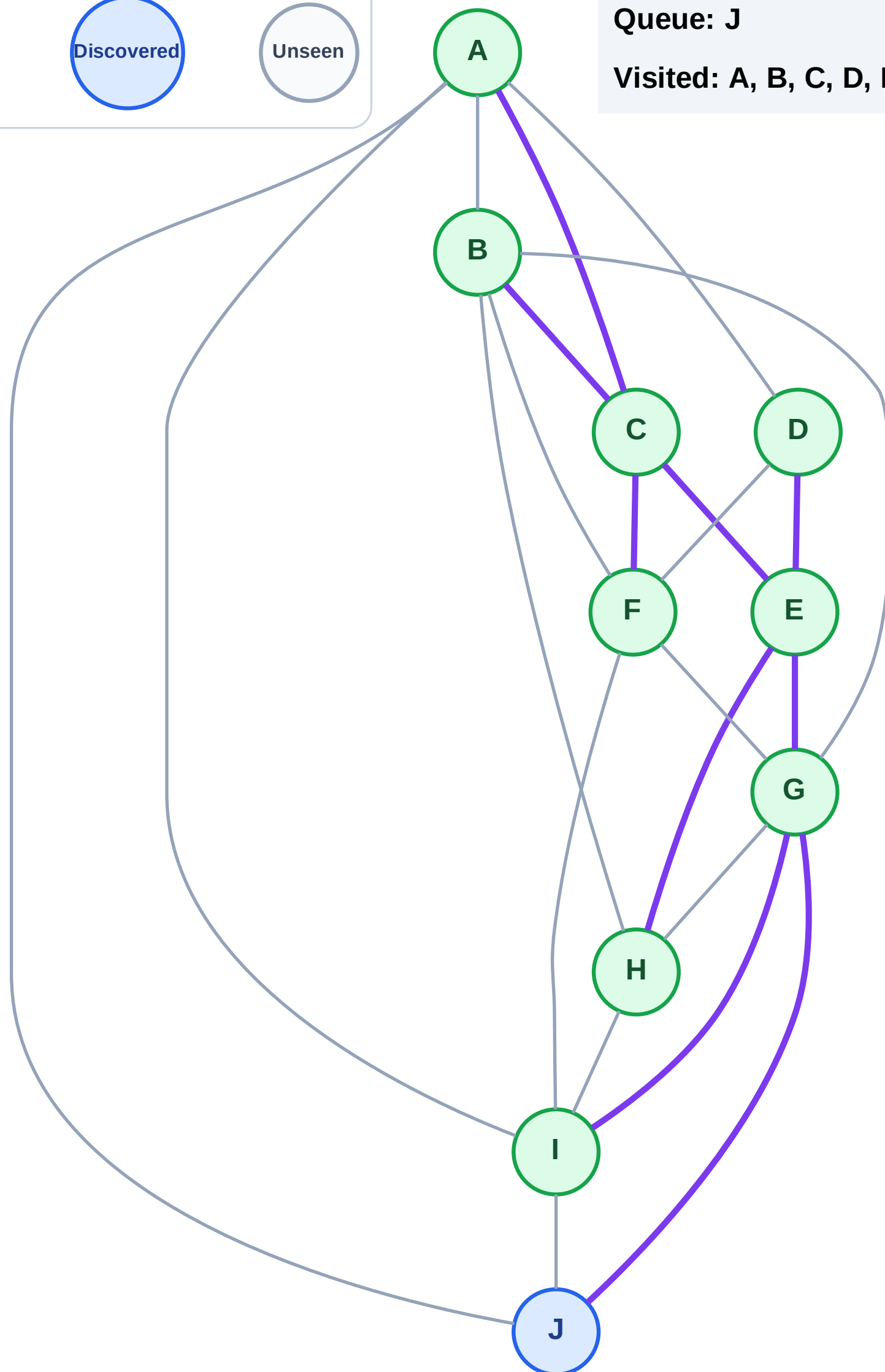
Step 19: No new neighbors from I

Legend



Queue: J

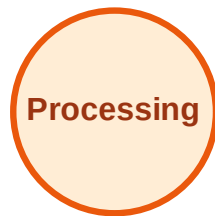
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

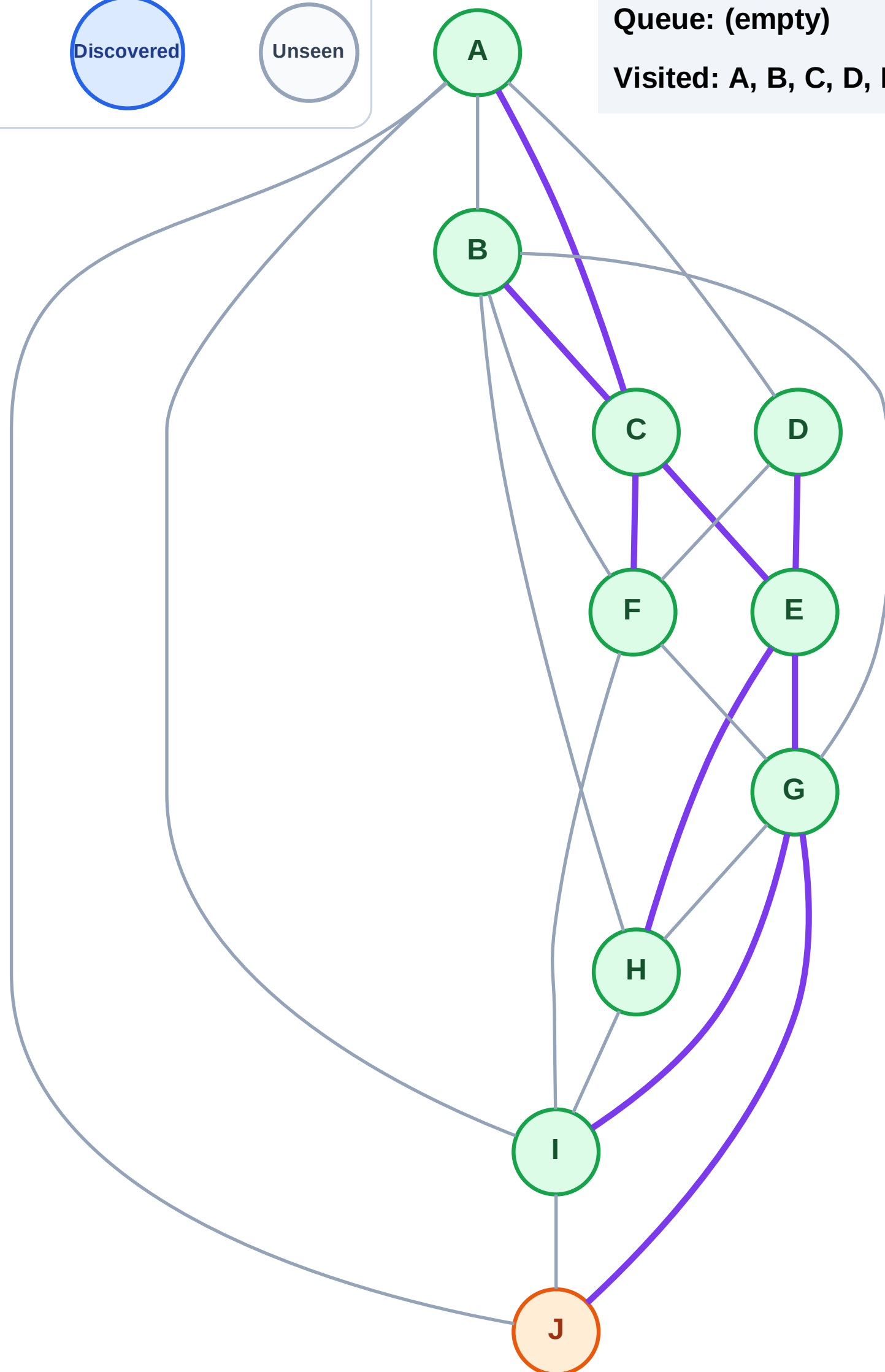
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

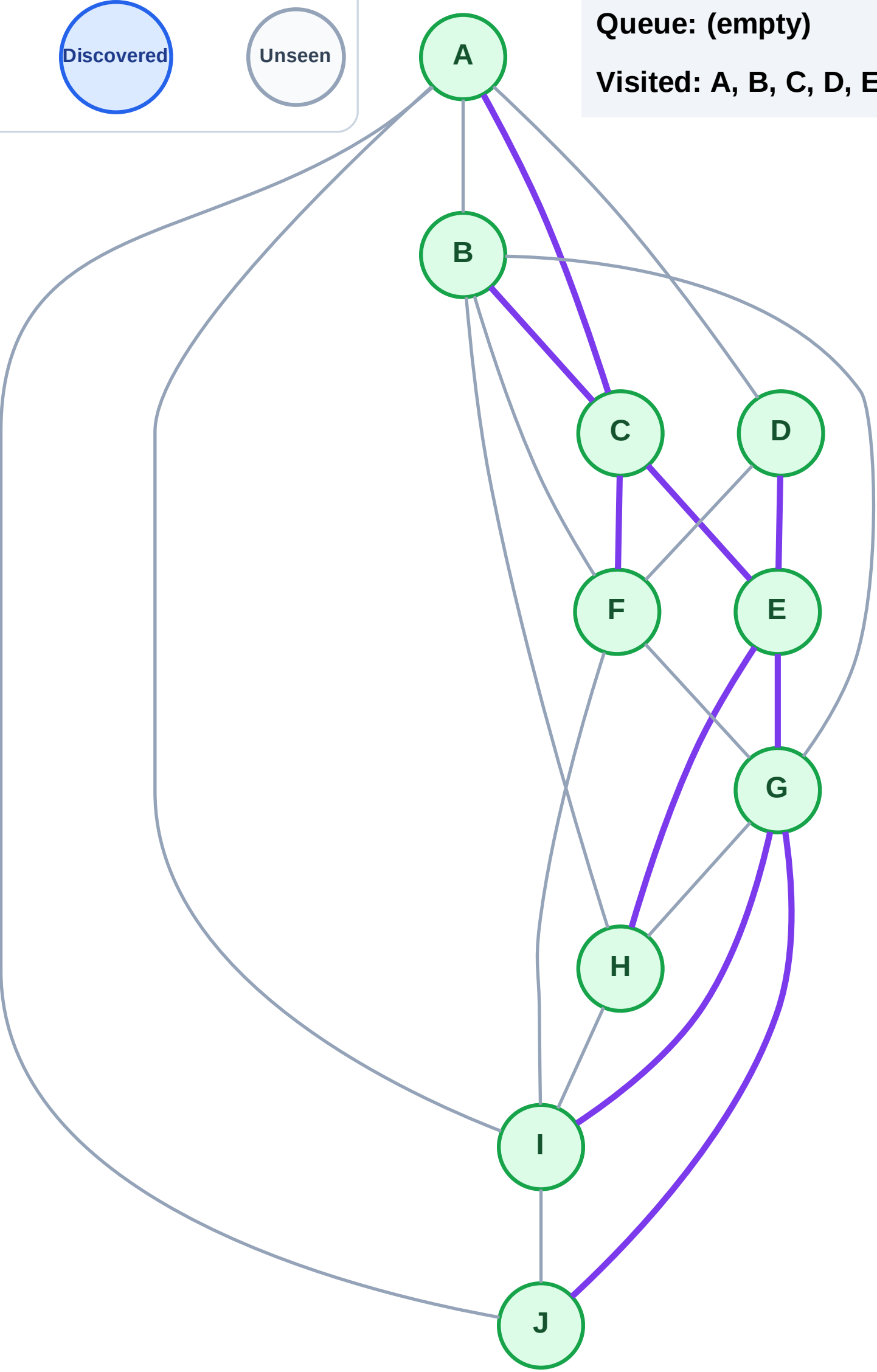
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

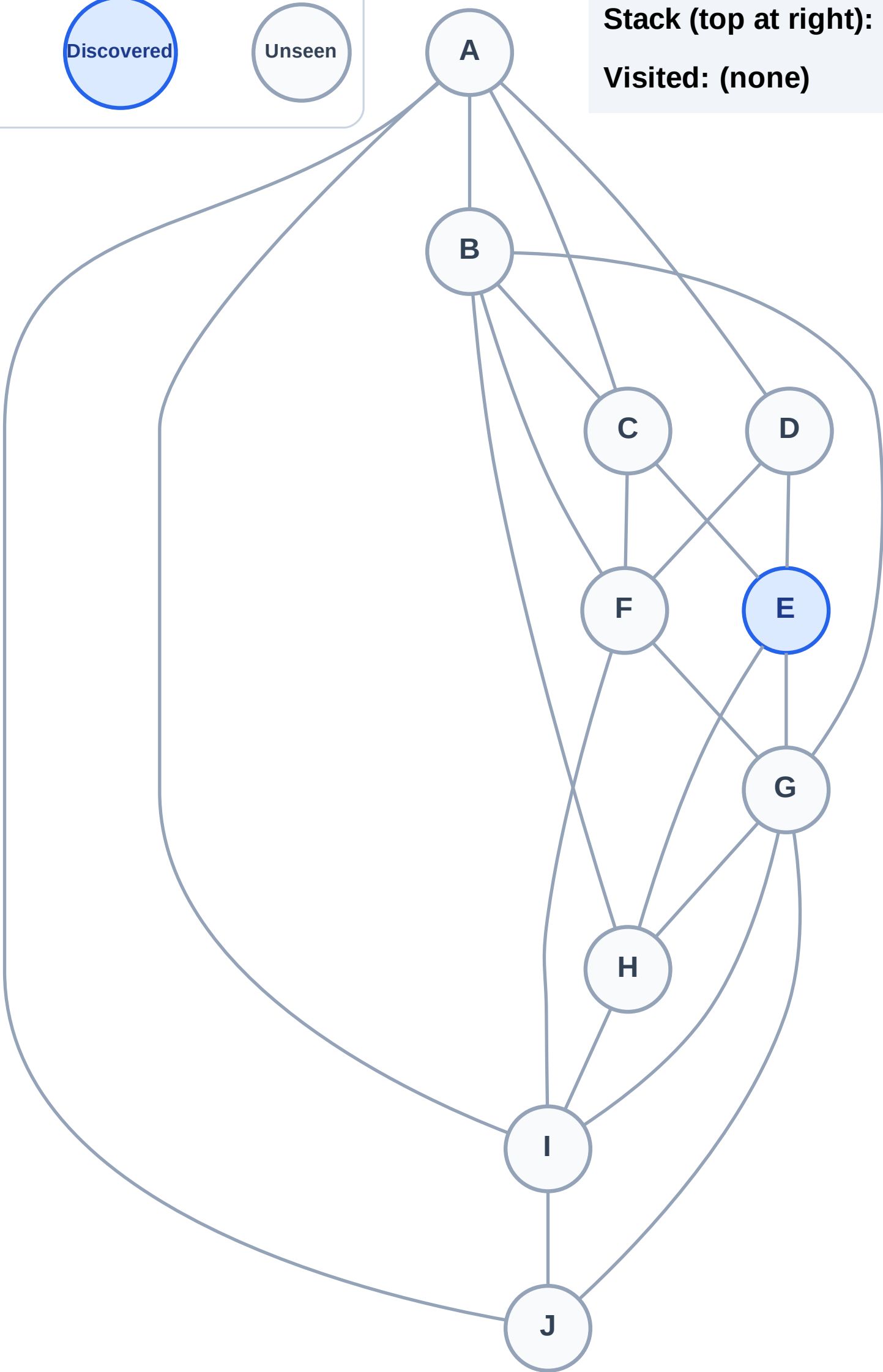
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

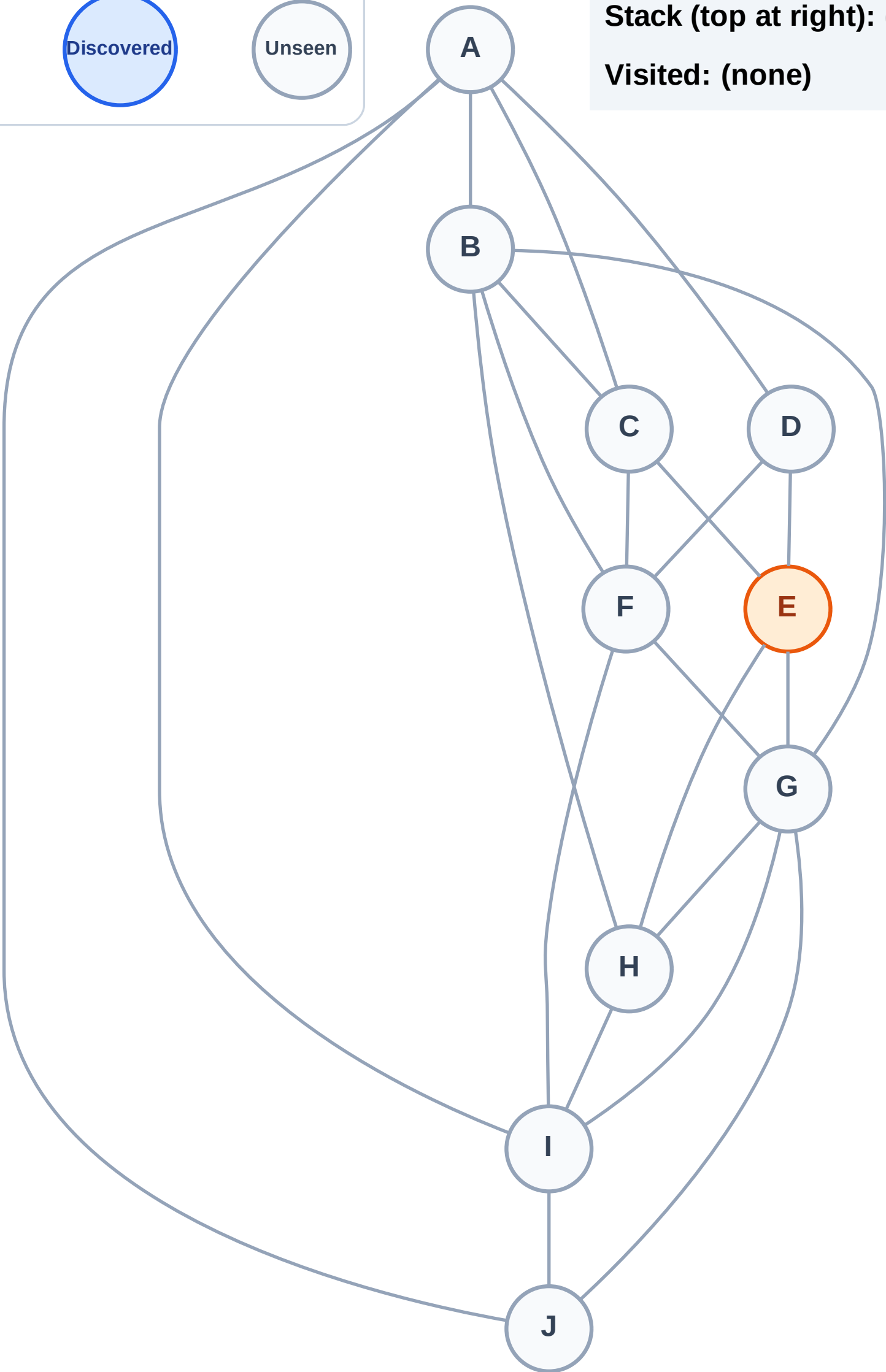
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, G, D, C from E

Legend

Complete

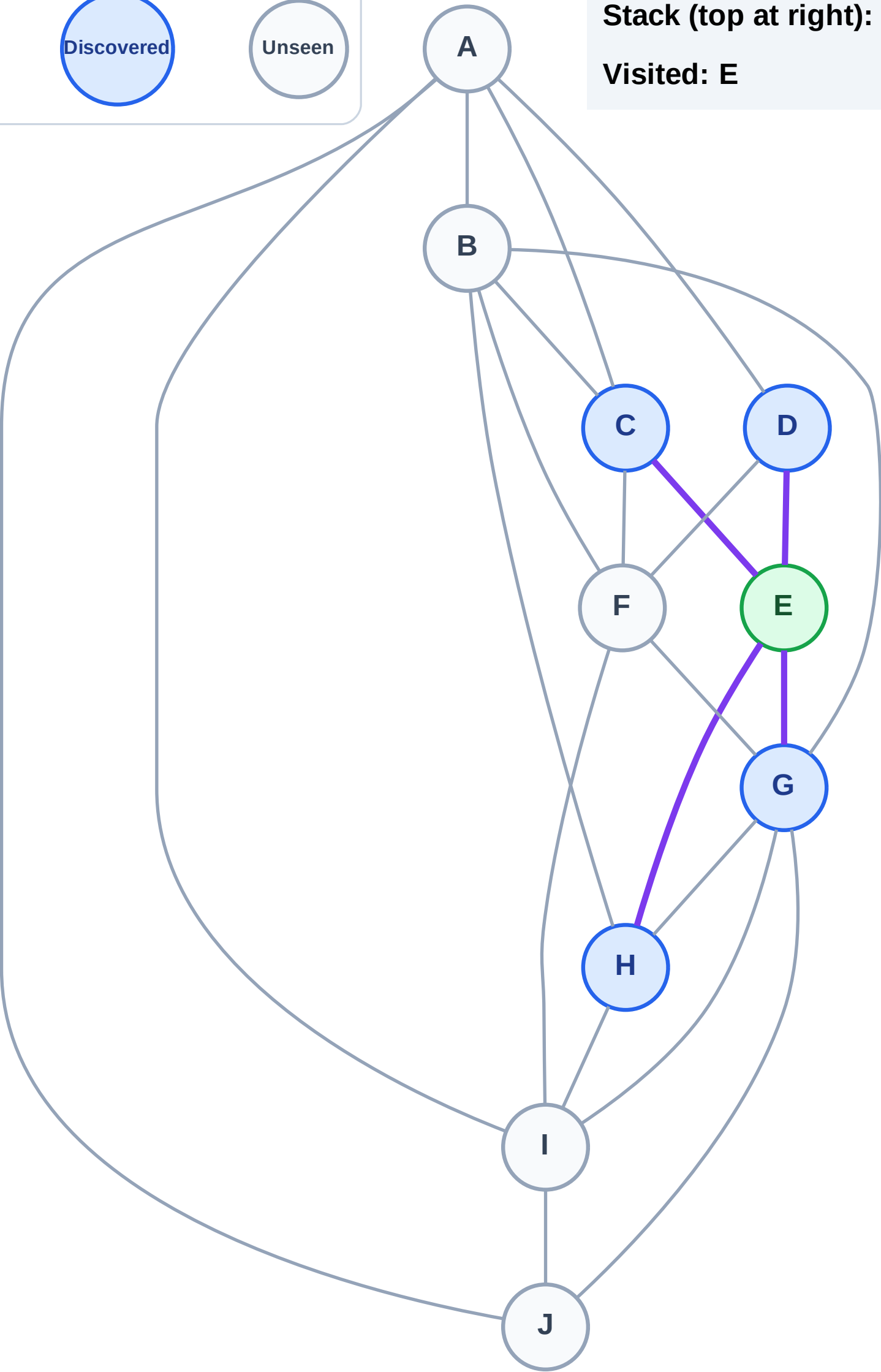
Processing

Discovered

Unseen

Stack (top at right): H | G | D | C

Visited: E



DFS Traversal

Step 4: Process node C

Legend

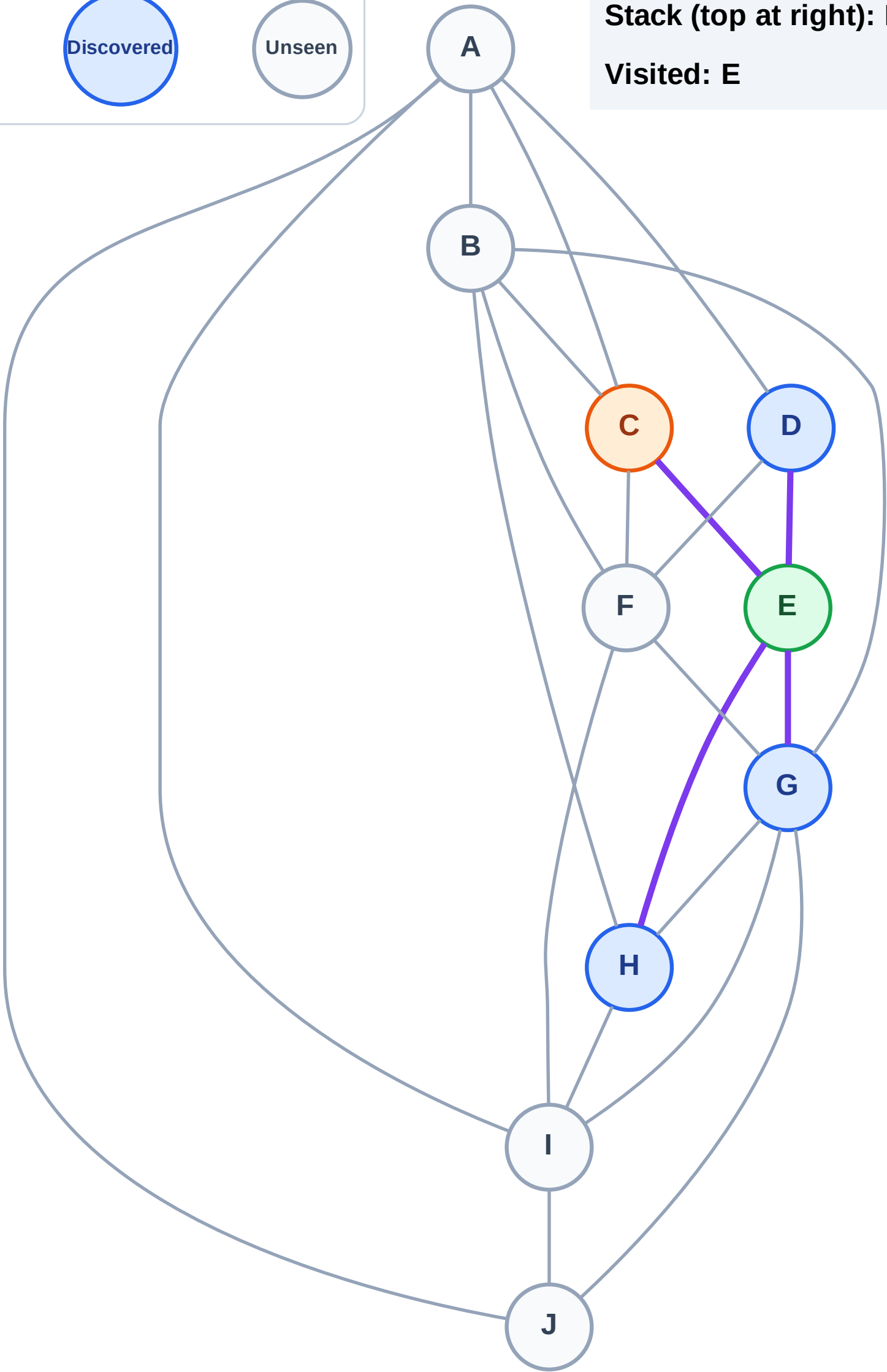
Complete

Processing

Discovered

Unseen

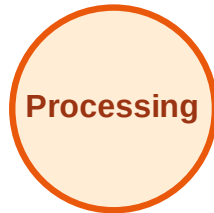
Stack (top at right): H | G | D
Visited: E



DFS Traversal

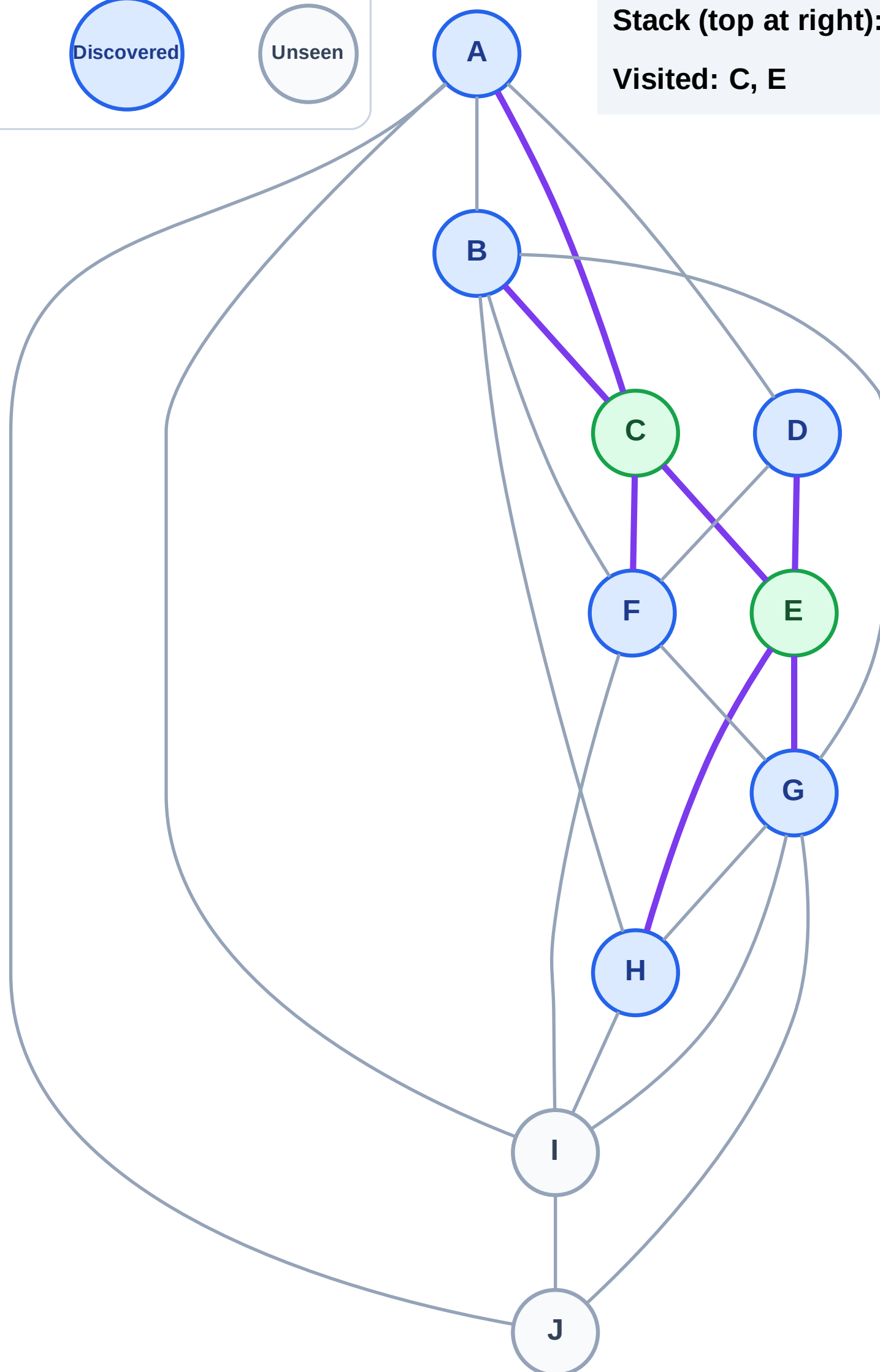
Step 5: Discover F, B, A from C

Legend



Stack (top at right): H | G | D | F | B | A

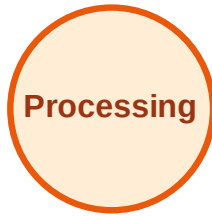
Visited: C, E



DFS Traversal

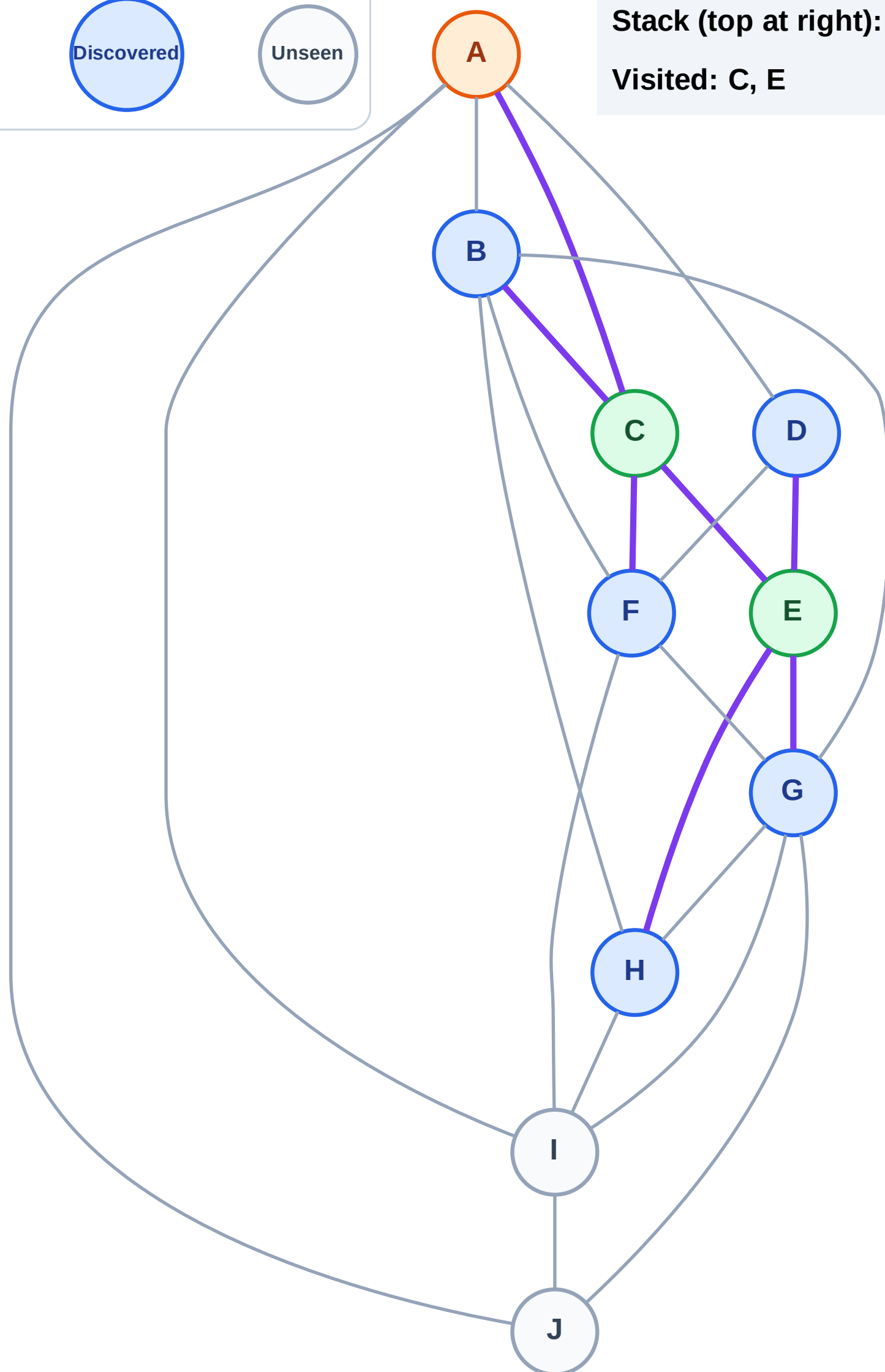
Step 6: Process node A

Legend



Stack (top at right): H | G | D | F | B

Visited: C, E



DFS Traversal

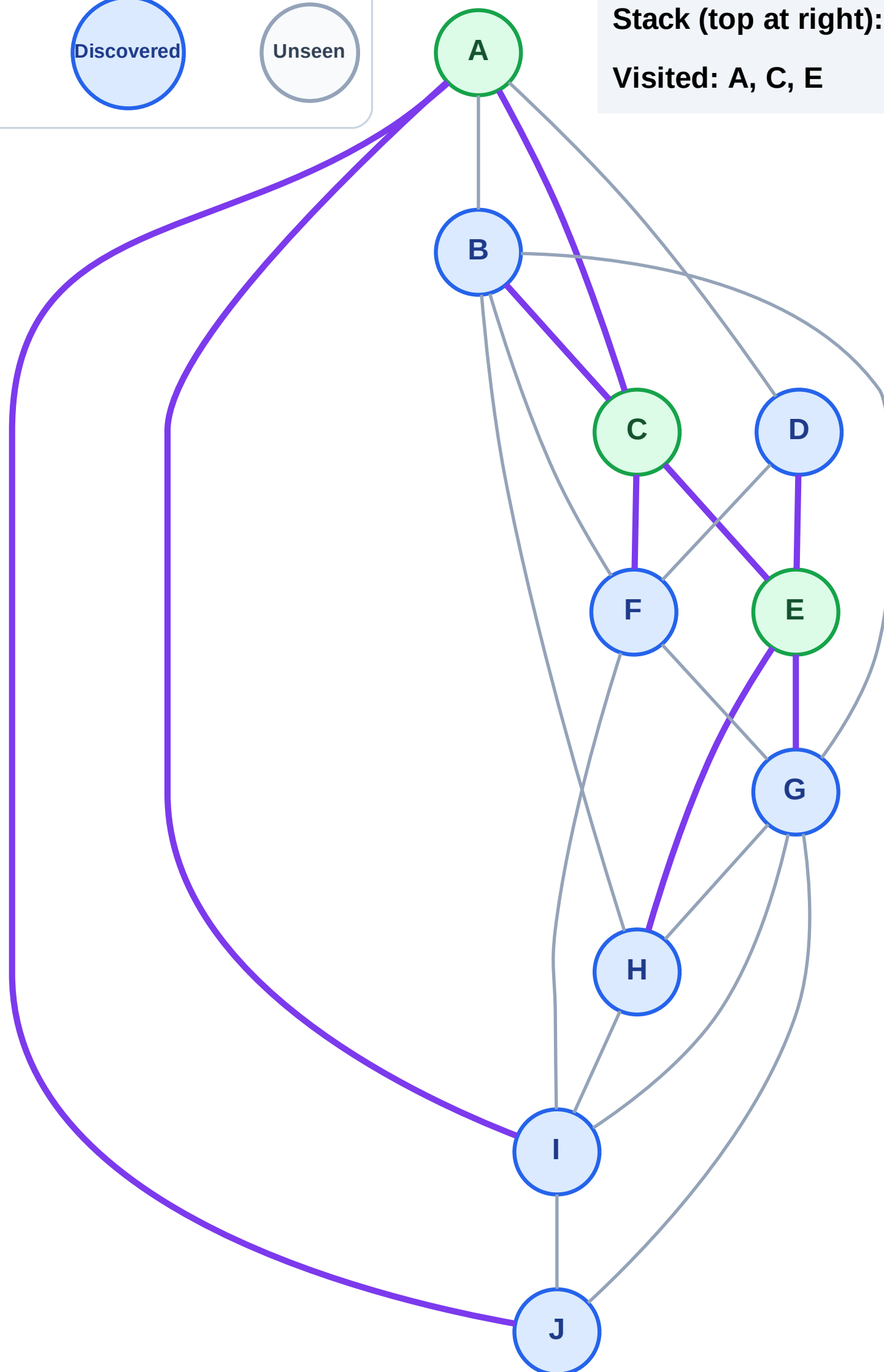
Step 7: Discover J, I from A

Legend



Stack (top at right): H | G | D | F | B | J | I

Visited: A, C, E



DFS Traversal

Step 8: Process node I

Legend

Complete

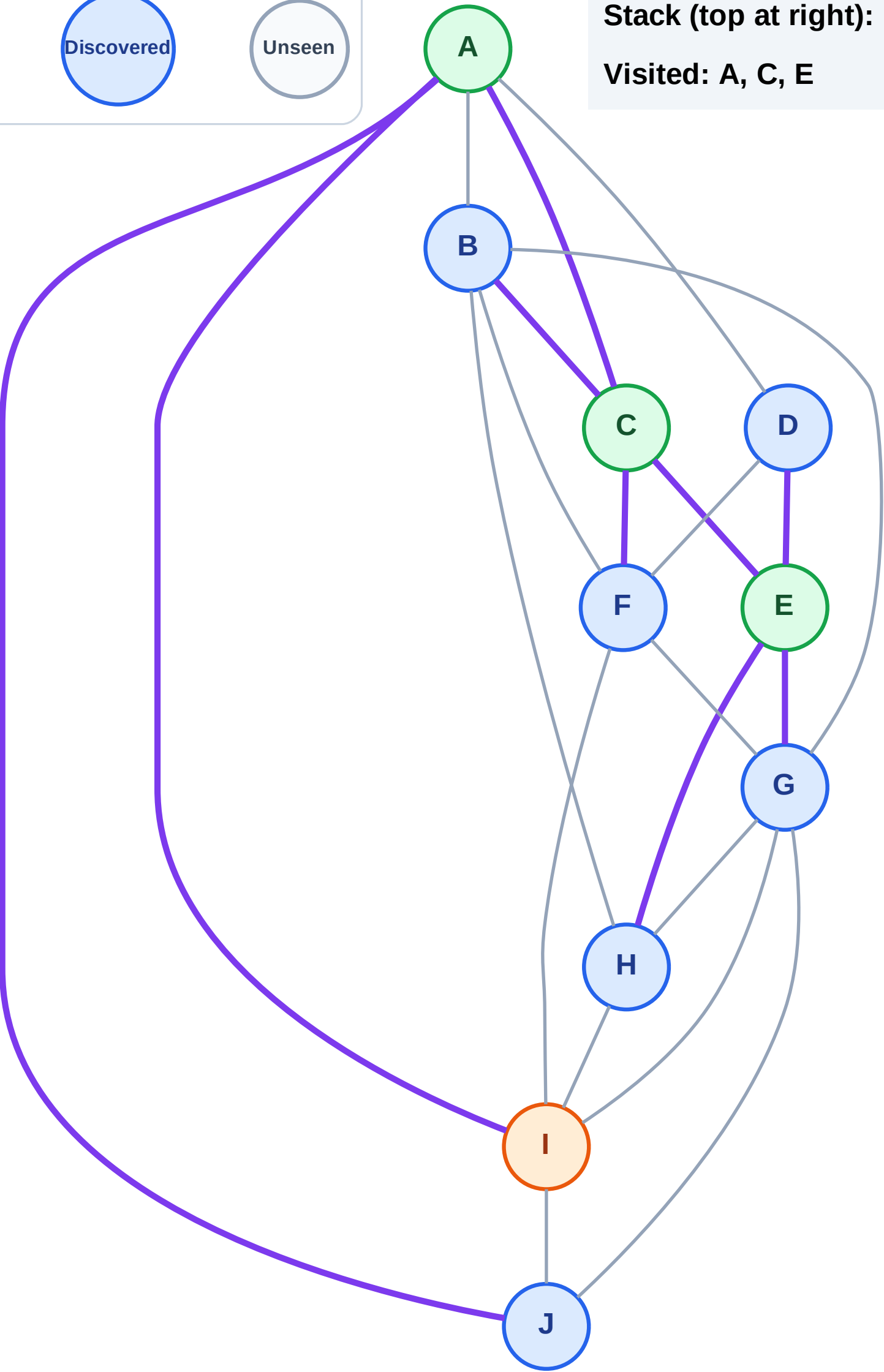
Processing

Discovered

Unseen

Stack (top at right): H | G | D | F | B | J

Visited: A, C, E



DFS Traversal

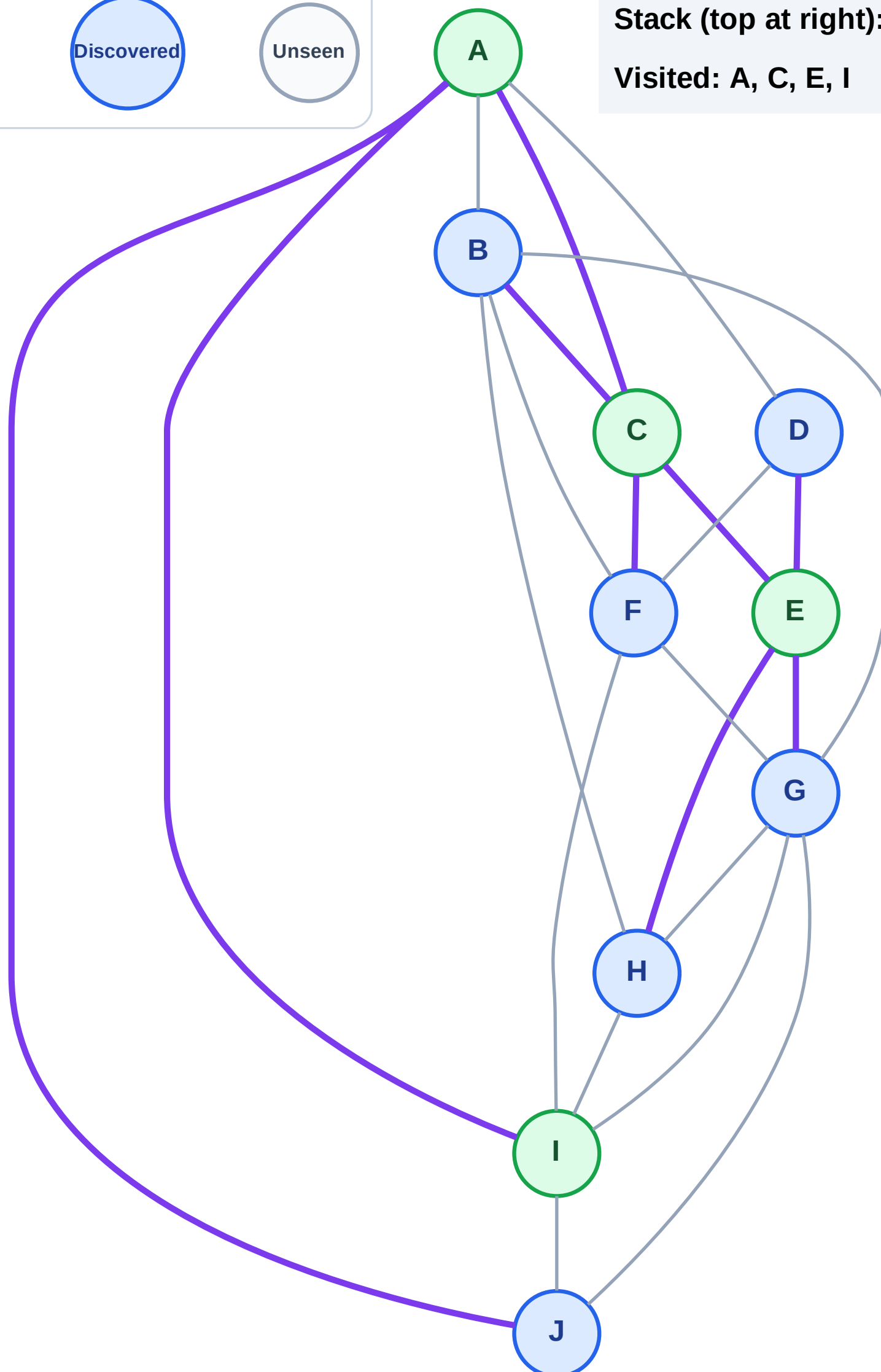
Step 9: No new neighbors from I

Legend



Stack (top at right): H | G | D | F | B | J

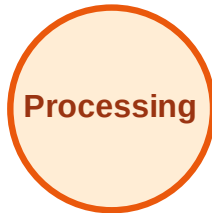
Visited: A, C, E, I



DFS Traversal

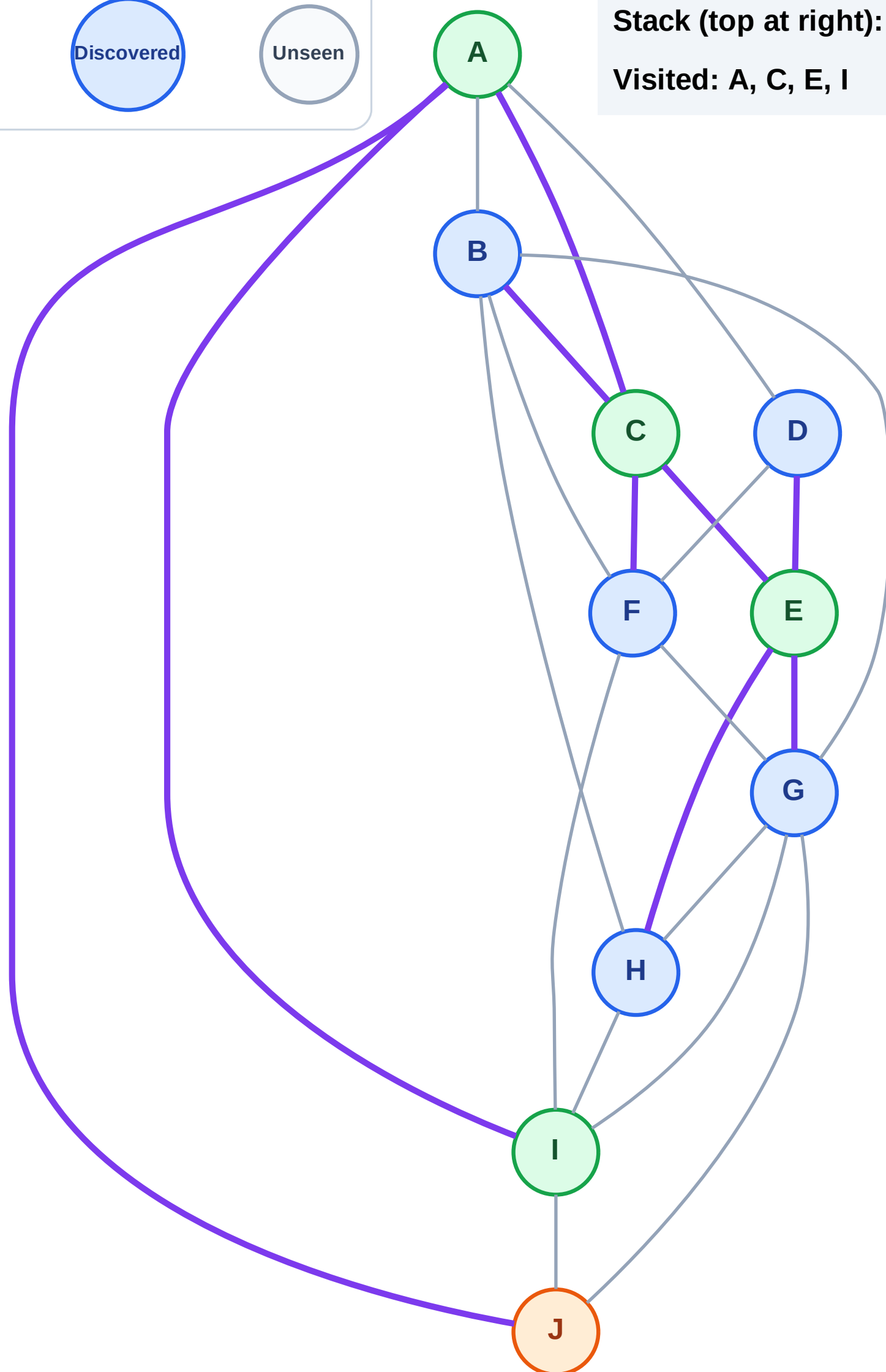
Step 10: Process node J

Legend



Stack (top at right): H | G | D | F | B

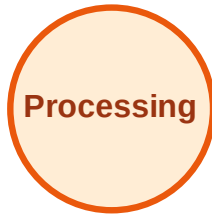
Visited: A, C, E, I



DFS Traversal

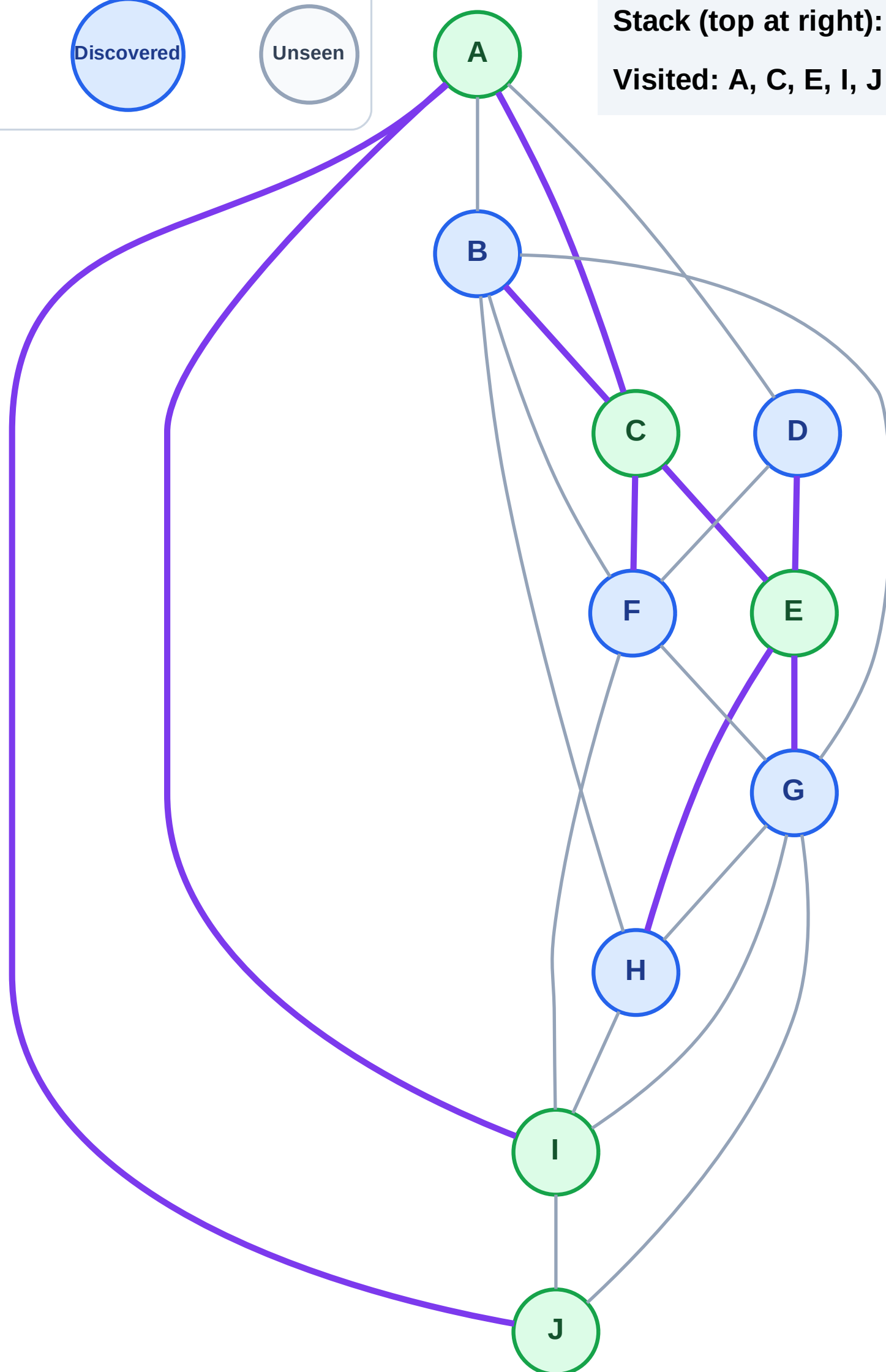
Step 11: No new neighbors from J

Legend



Stack (top at right): H | G | D | F | B

Visited: A, C, E, I, J



DFS Traversal

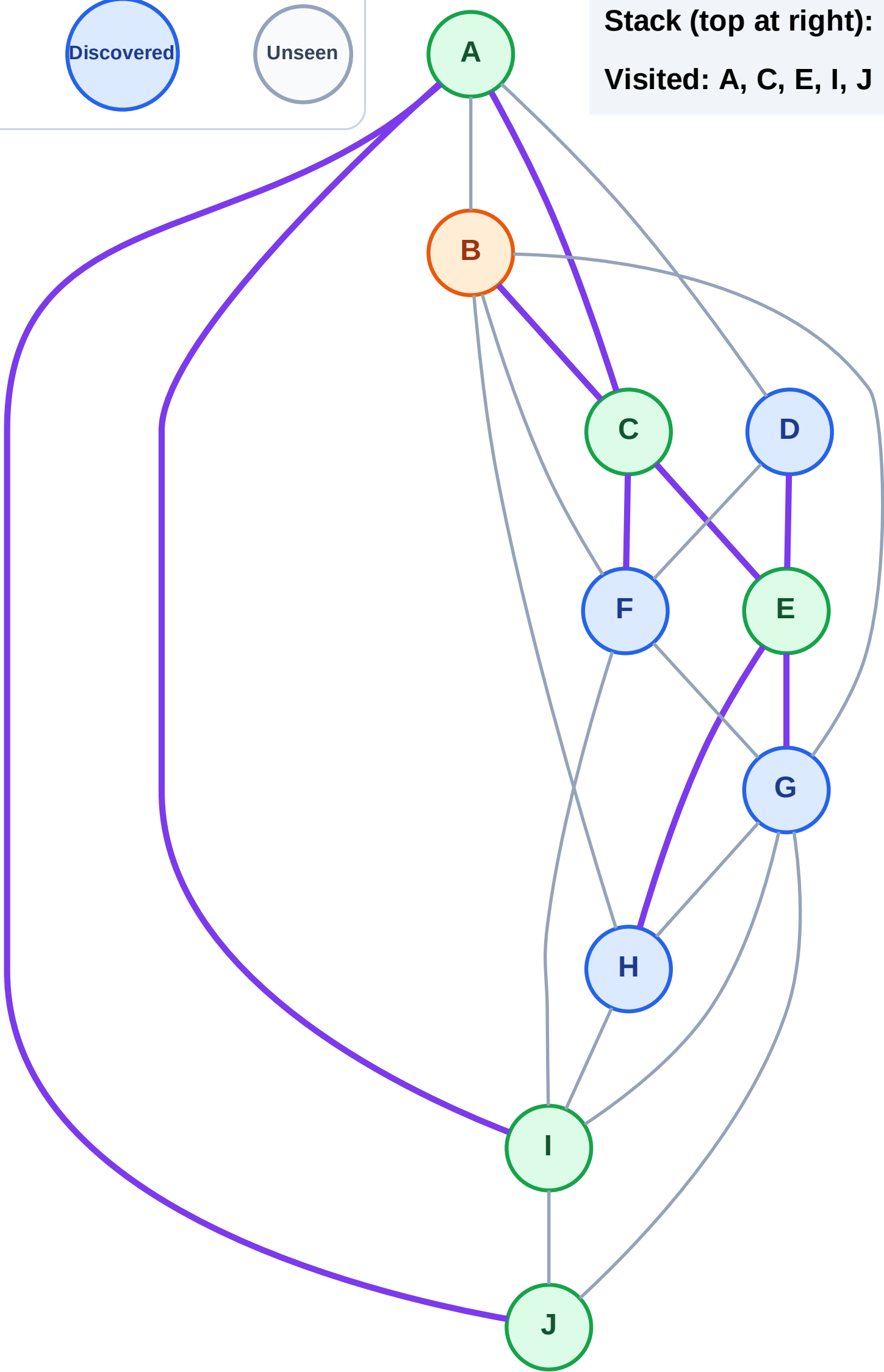
Step 12: Process node B

Legend



Stack (top at right): H | G | D | F

Visited: A, C, E, I, J



DFS Traversal

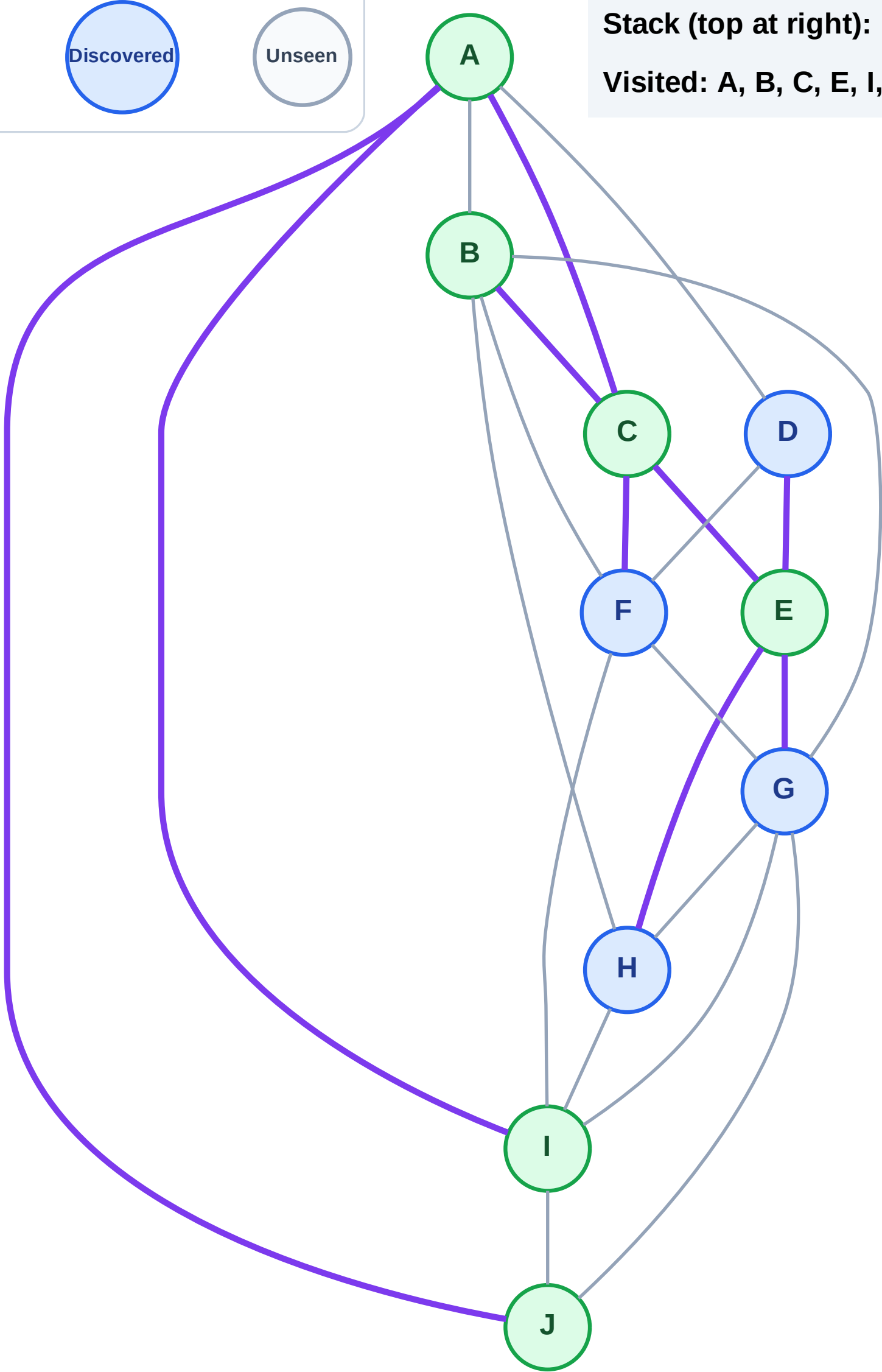
Step 13: No new neighbors from B

Legend



Stack (top at right): H | G | D | F

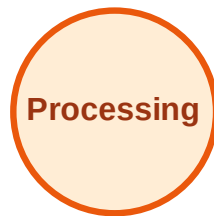
Visited: A, B, C, E, I, J



DFS Traversal

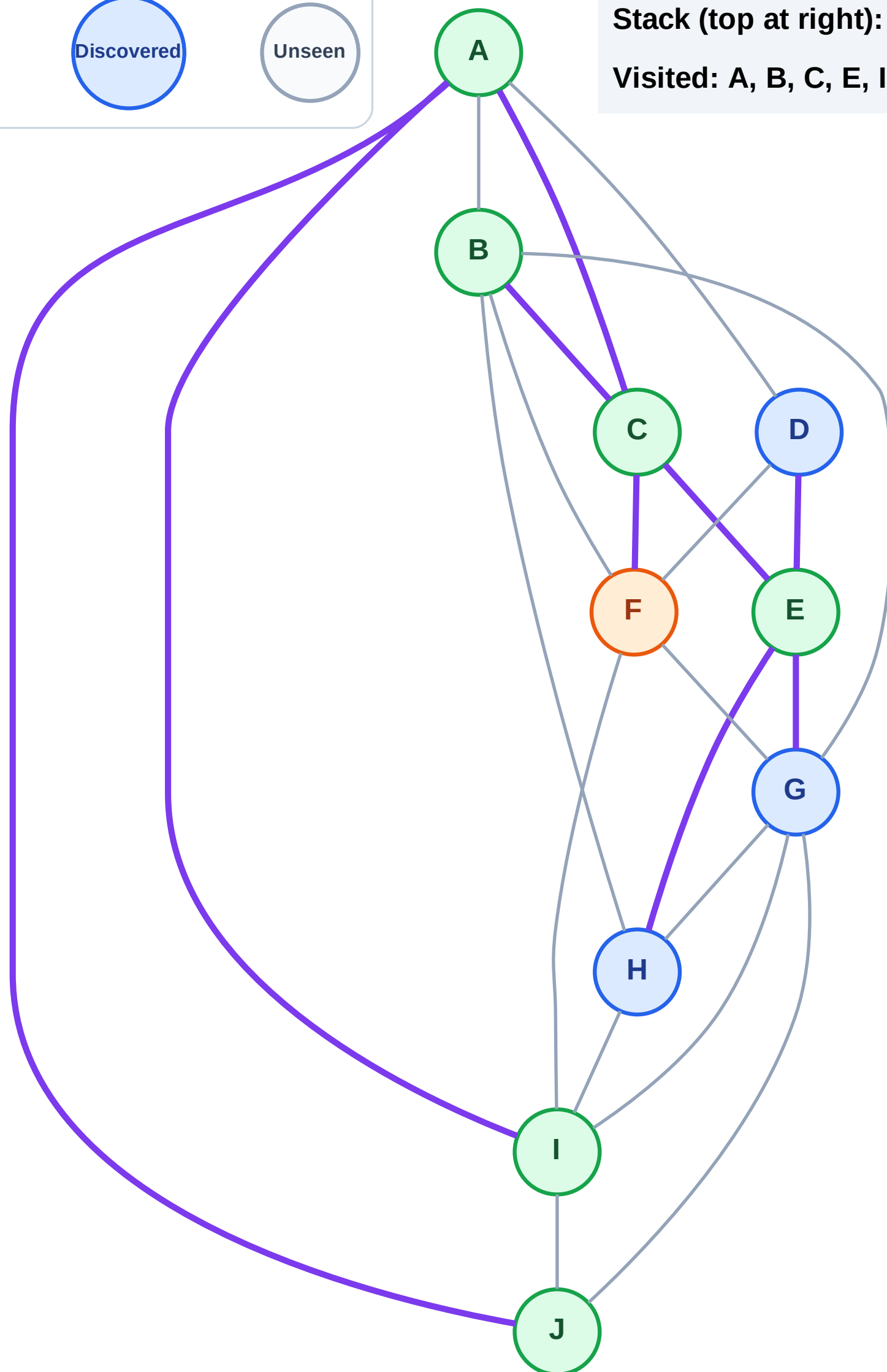
Step 14: Process node F

Legend



Stack (top at right): H | G | D

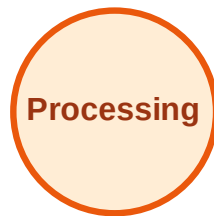
Visited: A, B, C, E, I, J



DFS Traversal

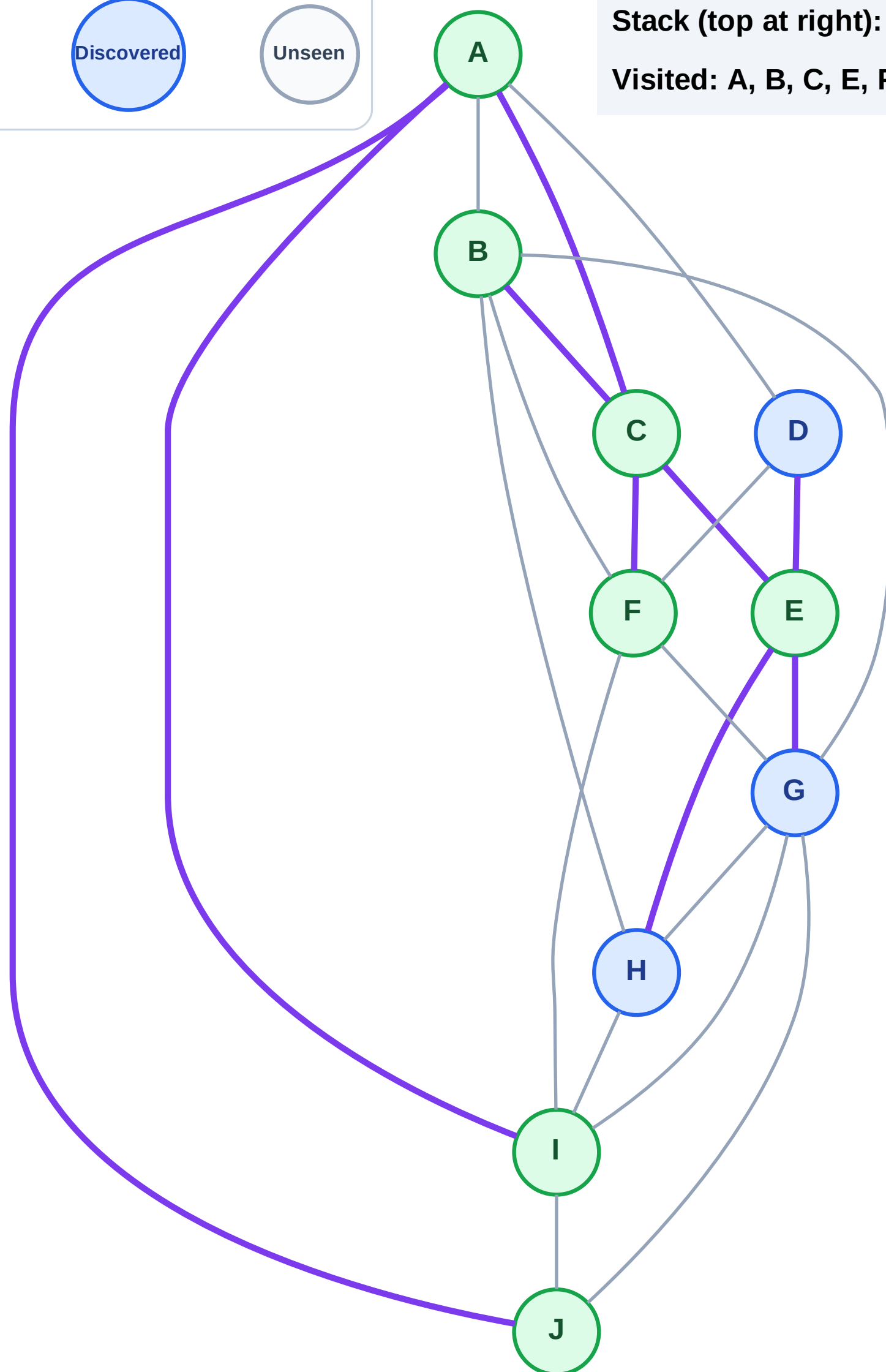
Step 15: No new neighbors from F

Legend



Stack (top at right): H | G | D

Visited: A, B, C, E, F, I, J



DFS Traversal

Step 16: Process node D

Legend

Complete

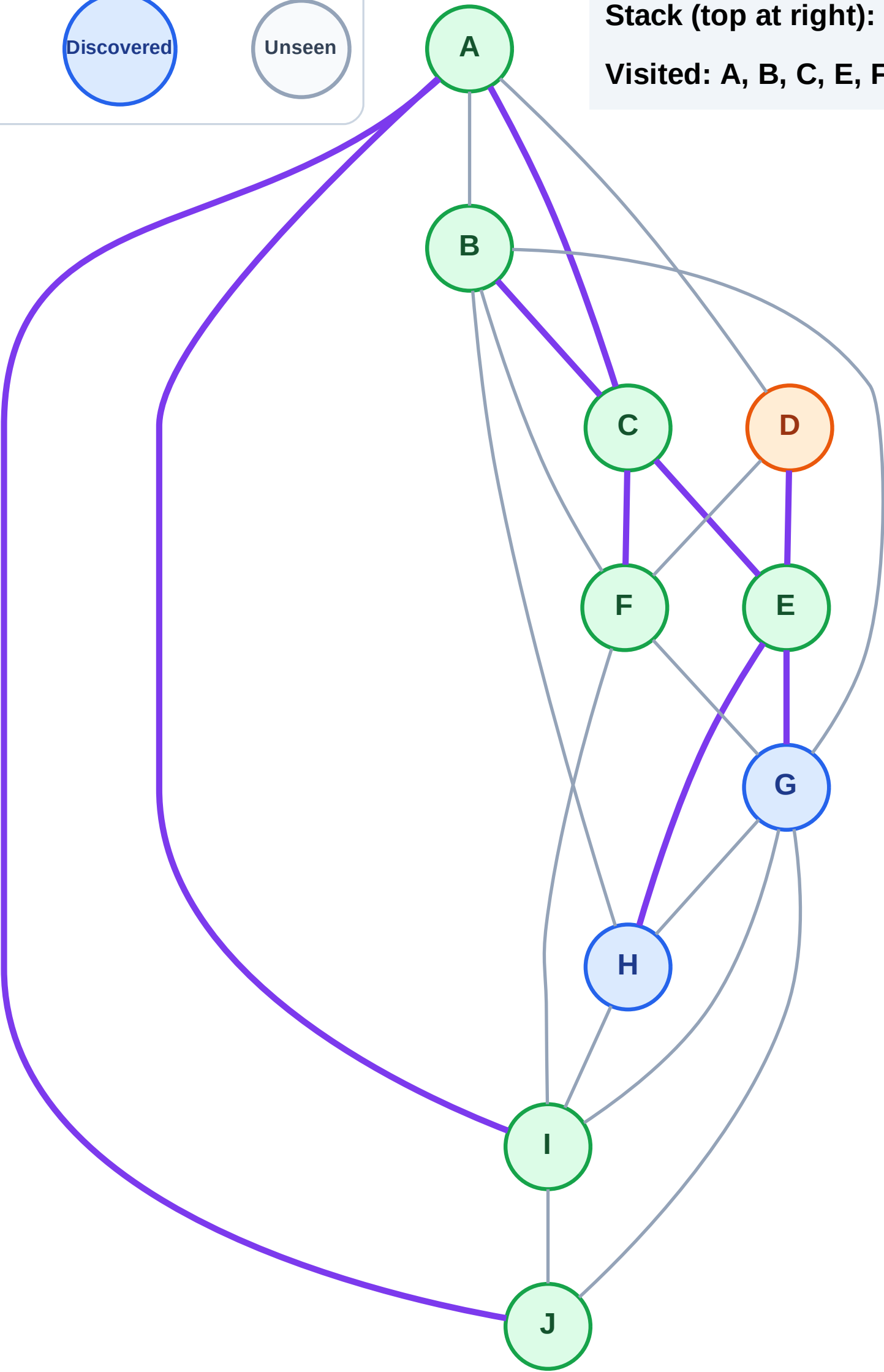
Processing

Discovered

Unseen

Stack (top at right): H | G

Visited: A, B, C, E, F, I, J



DFS Traversal

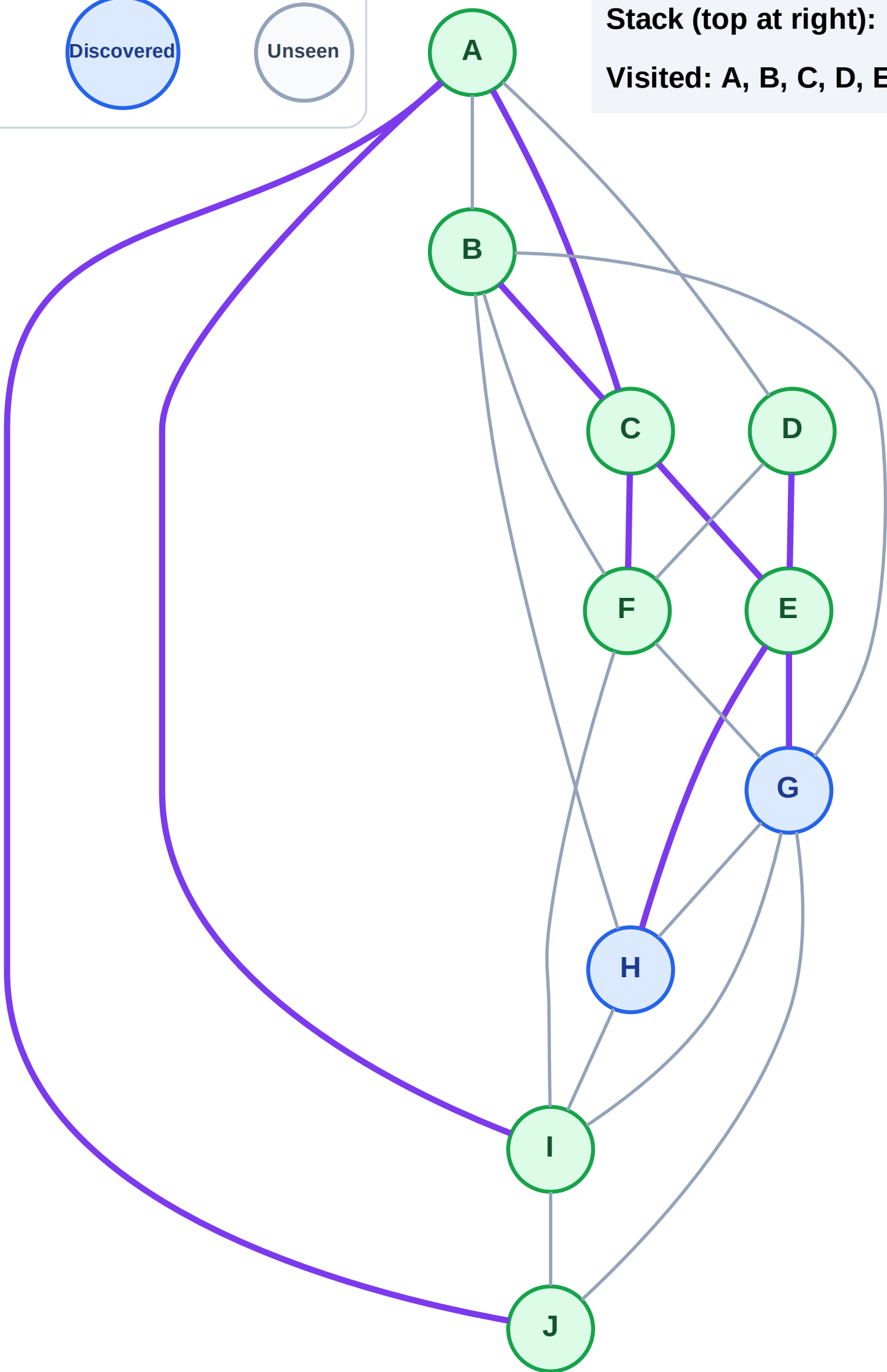
Step 17: No new neighbors from D

Legend



Stack (top at right): H | G

Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 18: Process node G

Legend

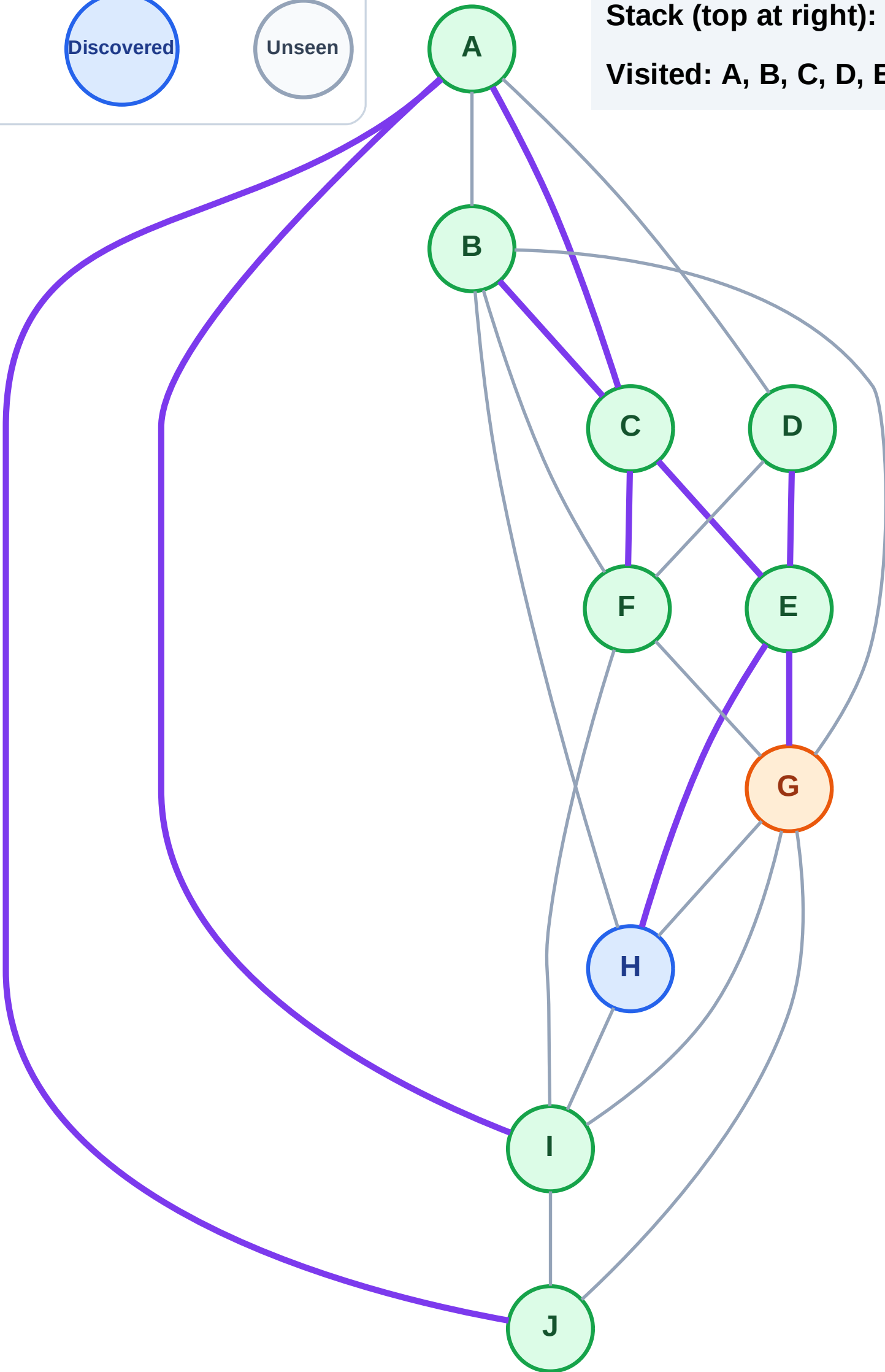
Complete

Processing

Discovered

Unseen

Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

Step 19: No new neighbors from G

Legend

Complete

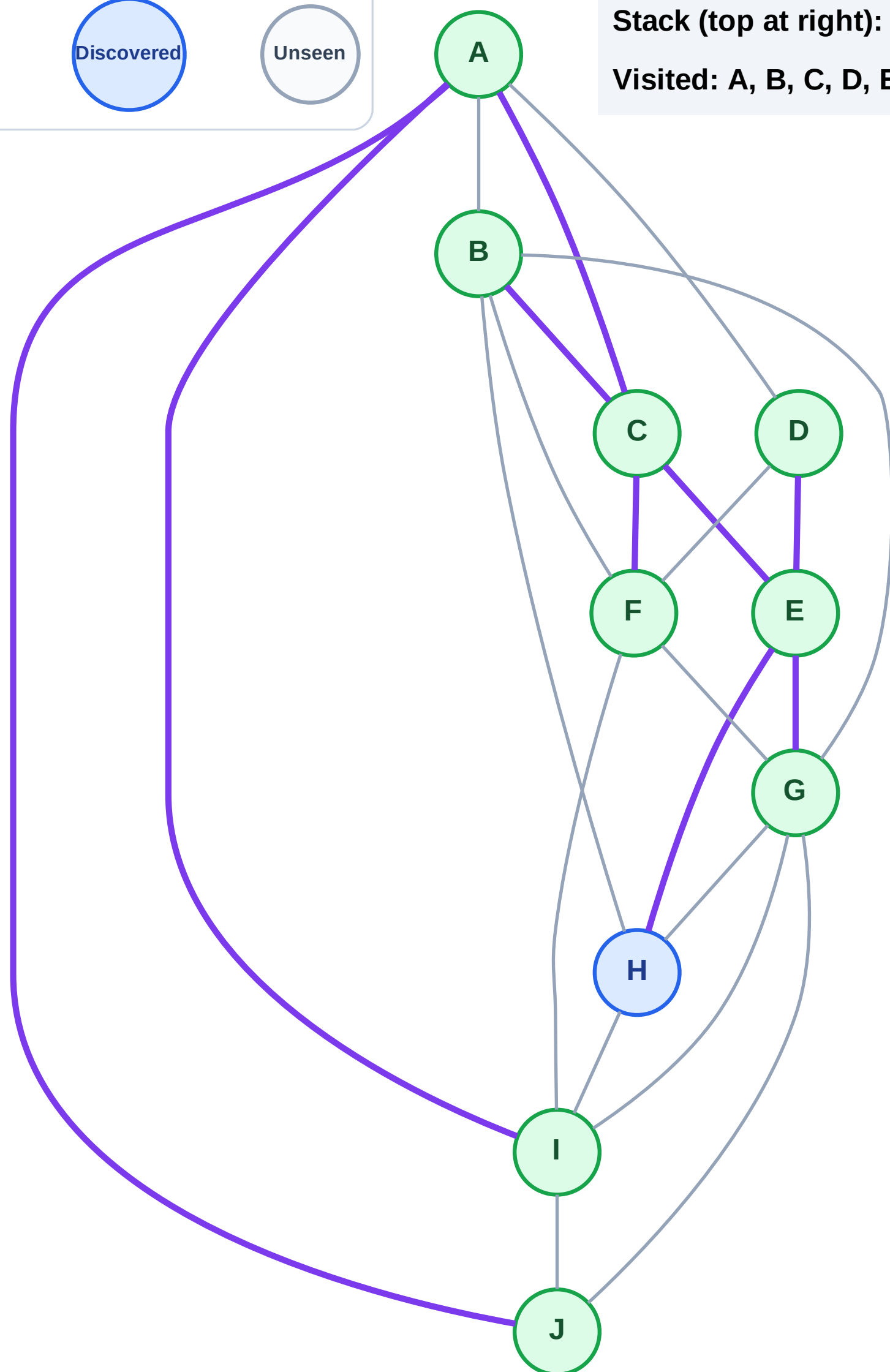
Processing

Discovered

Unseen

Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

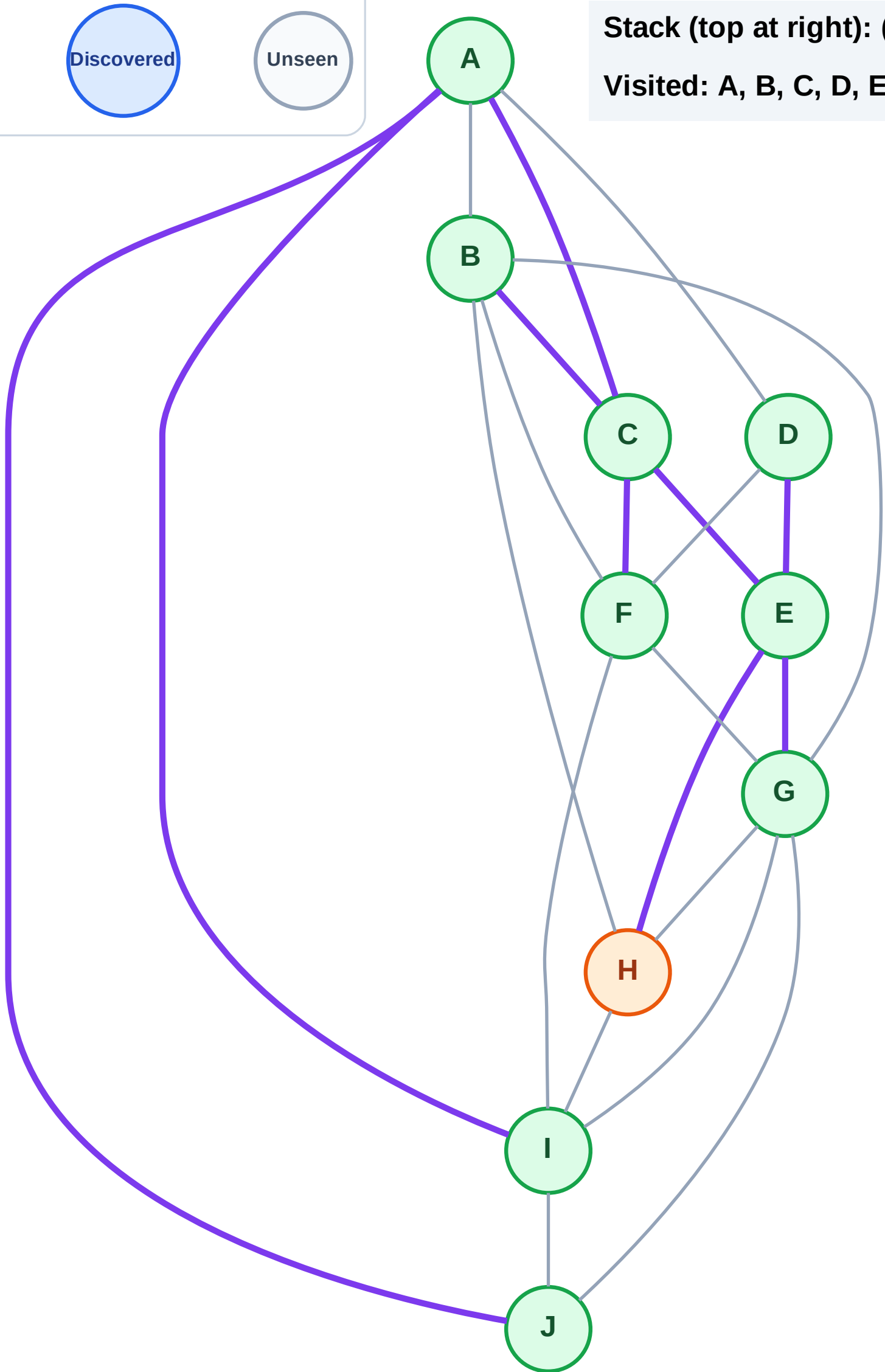
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

